A Reliability-Gated Cross-Domain Information Fusion Framework with Quantitatively Conditioned RAG-LLM for Personalized Rehabilitation Decision Support

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Abstract Older adults often present concurrent cognitive and motor impairments, but quantitative assessment and rehabilitation decision support are handled separately. This study proposes a reliability-gated cross-domain information fusion framework that connects cognitive-motor assessment with personalized rehabilitation decision support. A reliability-gated bidirectional domain-aware cross-attention medical Transformer (BDCMT) models heterogeneous demographic, cognitive, and motor measurements while preserving domain-specific representations and regulating cross-domain information exchange. A distributional ordinal loss further incorporates the ordered structure of impairment severity. Predicted grades, class-probability distributions, original measurements, deterministic scale summaries, demographic information, and attention-derived relevance cues are organized into a structured functional profile, which serves as an information-preserving interface for parallel retrieval from cognitive and motor rehabilitation knowledge bases. The structured profile, retrieved evidence, and predefined safety constraints are then integrated by a quantitatively conditioned retrieval-augmented large language model (QC-RAG-LLM) to generate personalized rehabilitation recommendations for clinician review. Across ten repeated runs on 977 community-dwelling older adults, BDCMT achieved mean accuracies of 0.8867 and 0.9036 for cognitive and motor impairment grading, respectively. In clinician evaluations of 40 stratified cases, the QC-RAG-LLM configuration received the highest ratings for safety, consensus, objectivity, reproducibility, and explainability. The framework provides a traceable pathway from quantitative assessment to evidence-informed personalized rehabilitation decision support.

Keywords Cross-domain information fusion; Cognitive-motor assessment; Ordinal learning; Retrieval-augmented generation; Large language models; Clinical decision support

1. Introduction

Population aging has increased the prevalence of cognitive impairment, mobility limitations, falls, and reduced independence among older adults. Cognitive and motor functions are closely interconnected in daily activities. Attention, executive control, and working memory contribute to gait regulation and balance, whereas declining mobility may further restrict physical activity, social participation, and cognitive stimulation. The Lancet committee identified insufficient physical activity, social isolation, sensory impairment, and vascular-related factors as important modifiable contributors associated with dementia risk [1]. These findings highlight the importance of considering cognitive and motor changes jointly rather than treating them as isolated functional domains when assessing decline in older adults.

This relationship is particularly evident in dual-task activities. When older adults are walking and performing cognitive tasks, gait control and cognitive processing compete for some of the shared attentional and executive resources, leading to cognitive-motor interference (CMI) [2,3]. Individuals with cognitive or motor dysfunction often exhibit increased dual-task costs, including reduced walking speed, impaired balance, and decreased cognitive performance, which are associated with higher fall risk and reduced independence [4]. Therefore, integrating cognitive and motor assessments may provide complementary information for characterizing functional status, particularly in individuals with impairments across both domains.

In community and rehabilitation settings, cognitive function is commonly assessed using standardized scales, while motor function is evaluated through measures such as fall history, the short physical performance battery (SPPB), grip strength, and the timed up and go (TUG) test. These assessments are clinically practical and interpretable, covering multiple functional aspects, including memory, attention, orientation, balance, muscle strength, gait, and mobility [5]. However, the challenge is not only the collection of these measurements but also the effective integration of heterogeneous information. Cognitive sub-items, motor performance indicators, demographic characteristics, and fall-related variables originate from different assessment domains and possess distinct clinical meanings. Simple aggregation into total scores or threshold-based categories may overlook interactions between domains, leaving their joint interpretation largely dependent on clinical expertise.

Machine learning methods can capture nonlinear dependencies in heterogeneous clinical tabular data [6,7], but cognitive-motor assessment poses a particular representation problem. Clinical variables retain predefined meanings associated with their source scales and functional domains, while their column order does not constitute a meaningful sequence. Mapping all variables into a single undifferentiated representation may obscure domain-specific characteristics. In contrast, modeling cognitive and motor measurements independently prevents potentially complementary information from being exchanged before prediction. The key challenge is therefore to preserve the structure of each functional domain while allowing controlled interaction between them. This balance is especially relevant for moderate-sized clinical datasets, where unconstrained cross-domain interactions may introduce noisy or irrelevant associations.

In addition, impairment severity exhibits an inherent ordinal structure. In this study, cognitive and motor status are represented using four severity levels ranging from normal to severe impairment. These categories are ordered rather than independent. For example, predicting severe impairment as normal represents a larger clinical deviation than predicting it as moderate impairment. Conventional cross-entropy loss treats classes as nominal categories and does not explicitly consider such severity relationships [8,9]. Therefore, cognitive-motor assessment requires not only effective cross-domain representation learning but also an optimization strategy that reflects the ordered characteristics of clinical severity.

However, accurate assessment and classification are only one part of the rehabilitation process. Most medical AI systems tend to focus solely on labels, probabilities, or risk scores [10,11]. Rehabilitation plans require information about the measurements underlying those predictions, scale-specific functional deficits, safety considerations, and relevant clinical evidence. Reducing a multidimensional assessment to a single prediction can therefore discard information that remains useful after the prediction stage. This issue is particularly relevant when cognitive and motor impairments coexist, as rehabilitation planning may involve cognitive training, physical activity, strength and mobility exercises, fall prevention, supervision, and follow-up adjustment. Preserving clinically relevant quantitative information beyond the predicted grade is thus important for connecting assessment with subsequent decision support.

Large language models (LLMs) provide a practical approach to organizing heterogeneous clinical information into easily understandable decision support content [12,13]. However, directly applying them to rehabilitation programs still presents some challenges. The generated recommendations may lack supporting evidence, omit relevant assessment results, or the suggested interventions may not match the individual's functional status. Retrieval-augmented generation (RAG) can reduce some of these limitations by grounding generation in clinical guidelines, expert consensus, and rehabilitation protocols [14,15]. However, existing clinical RAG approaches often rely on textual queries or concise case descriptions. Such inputs may not preserve quantitative information obtained from upstream assessments, including predicted grades, probability distributions, original measurements, and model-derived relevance cues. Consequently, retrieved information may be generally relevant while remaining insufficiently tailored to the individual's measured functional characteristics.

To address these challenges, this study develops a reliability-gated cross-domain information fusion framework that maintains quantitative cognitive-motor information throughout assessment, evidence retrieval, and rehabilitation decision support. A reliability-gated bidirectional domain-aware cross-attention medical Transformer (BDCMT) is proposed to model heterogeneous demographic, cognitive, and motor measurements. The model preserves domain-specific representations while using reliability-gated bidirectional interactions to capture complementary information between cognitive and motor domains. A distributional ordinal loss is further introduced to incorporate the ordered characteristics of impairment severity into model optimization.

The predicted assessment results were then integrated with class-probability distributions, original measurements, deterministic scale summaries, demographic information, and attention-derived relevance cues into a structured functional profile. This profile serves as an interface between quantitative assessment and rehabilitation decision support. It allows for parallel retrieval of information from independent cognitive and motor rehabilitation knowledge bases, and subsequently combines the retrieved evidence with safety constraints through a quantitatively conditioned retrieval-augmented large language model (QC-RAG-LLM). By integrating the structured profile, retrieved evidence, and safety constraints, the proposed framework supports personalized rehabilitation recommendations for clinician review. The main contributions of this study are as follows:

- A reliability-gated cross-domain representation learning model (BDCMT) is proposed for cognitive-motor assessment. Domain-specific cognitive and motor representations are preserved through a dual-stream architecture, while bidirectional cross-attention with adaptive gating is used to regulate cross-domain information exchange.
- A distributional ordinal loss is introduced to incorporate the ordered relationships among impairment severity levels into model optimization. Both grade proximity and cumulative distribution consistency are considered to reduce large-distance grading errors.
- A structured assessment-to-decision interface is constructed by integrating predicted grades, class-probability distributions, original measurements, deterministic scale summaries, demographic information, and attention-derived relevance cues into a structured functional profile. Quantitative assessment information is thereby retained for downstream evidence retrieval and decision support.
- A quantitatively conditioned RAG-LLM framework is developed to connect the structured functional profile with parallel cognitive and motor evidence retrieval. Retrieved evidence and predefined safety constraints are incorporated into the generation process to support personalized rehabilitation recommendations for clinician review.

The remainder of this paper is organized as follows. Section 2 reviews related work. Section 3 presents the proposed framework, including BDCMT, the distributional ordinal loss, the structured assessment-to-decision interface, and QC-RAG-LLM. Section 4 reports the experimental results. Section 5 discusses the findings and limitations. Section 6 concludes the paper.

2. Related Work

2.1 Deep Learning for Clinical Tabular Data

Clinical assessments are commonly represented as structured tabular data, including demographic characteristics, scale scores, physical measurements, laboratory indicators, and categorical history variables [16]. Unlike images or free text, these data are typically low-dimensional and heterogeneous, and each variable has a predefined meaning determined by its clinical source. Tree-based methods remain competitive for many tabular prediction tasks, particularly when the available sample size is limited [17]. Deep neural

models, however, offer greater flexibility for representation learning and for extending tabular prediction to multi-task or downstream decision-support settings.

A number of neural architectures have been developed for tabular data. TabNet [18] uses sequential attention to select informative variables across multiple decision steps, whereas recent MLP-based approaches such as TabM [19] improve predictive performance through parameter-efficient ensembling. Self-supervised methods have also been explored to make better use of limited tabular data. VIME [20] learns representations through feature corruption, mask estimation, and reconstruction, while SCARF [21] adopts contrastive learning based on randomly corrupted features. Other approaches address transferability or multi-task prediction. TransTab [22] develops transferable representations across heterogeneous tables, whereas MMoE [23] uses shared experts with task-specific gating to model related prediction tasks.

These studies provide useful tools for clinical tabular modeling, but cognitive-motor assessment introduces additional structure. Cognitive and motor measurements come from clinically distinct yet related functional domains. Combining all variables in a single representation may weaken this distinction, while modeling the domains independently limits information exchange between them. In addition, impairment severity is ordinal rather than purely categorical. A cognitive-motor assessment model therefore needs to retain domain-specific information, capture cross-domain relationships, and account for the ordered nature of severity levels. These limitations indicate that clinical cognitive-motor assessment can be viewed as a cross-domain information fusion problem, where domain-specific representations and controlled interactions are both required.

2.2 Cognitive-Motor Assessment and Rehabilitation

Cognitive and motor functions are closely interconnected in older adults and jointly influence functional independence. Cognitive processes, including attention, executive function, and working memory, contribute to gait regulation, postural control, and adaptation to environmental changes. Conversely, reduced mobility may limit physical activity and social participation, which are associated with cognitive health. Dual-task gait studies have demonstrated relationships between cognitive impairment and alterations in gait speed, balance, and dual-task performance, indicating that motor measurements can provide complementary information for assessing functional status [24,25].

The same relationship is relevant to rehabilitation. Cognitive-motor dual-task training combines cognitive exercises with gait, balance, strength, or coordination tasks and has been used to address the interaction between cognitive control and motor performance [26]. Fall-prevention guidelines likewise emphasize multifactorial assessment rather than a single functional measure, considering cognition together with gait, balance, muscle strength, medication, environmental risks, and fall history [27]. These findings suggest that rehabilitation decision support for older adults should consider cognitive and motor characteristics jointly rather than relying on isolated functional indicators.

In routine practice, cognitive function is commonly evaluated using standardized scales, whereas motor

function is characterized using measures such as fall history, the short physical performance battery (SPPB), grip strength, and the timed up and go (TUG) test. These measurements are clinically interpretable and suitable for community-based assessment, but their integration remains challenging. Cognitive sub-items, physical performance indicators, demographic characteristics, and fall-related variables originate from different functional domains and contain heterogeneous information. Simple feature concatenation or independent prediction models may capture partial relationships, but they do not explicitly preserve domain-specific structures while controlling information exchange between cognitive and motor domains. Therefore, cognitive-motor assessment requires information fusion approaches that can retain domain characteristics and exploit complementary interactions between heterogeneous functional domains.

2.3 Retrieval-Augmented Language Models for Clinical Decision Support

Large language models (LLMs) have demonstrated strong language organization capabilities in medical question answering, clinical text generation, and health dialogues [28,29]. However, their application in clinical decision support has raised some concerns, such as unsubstantiated statements, incomplete recommendations, and inconsistencies with clinical guidelines. These issues are particularly important in rehabilitation therapy, as inappropriate exercise intensity, movement selection, or safety advice can have adverse consequences for older adults.

Retrieval-augmented generation (RAG) addresses part of this problem by retrieving relevant external information and supplying it to the language model during generation. Clinical RAG systems have used guidelines, expert consensus, and other medical resources to improve factual grounding and the traceability of generated responses [30,31]. Related approaches have also been examined in surgical decision support, medical fitness assessment, and other healthcare applications [32,33].

Most clinical RAG procedures are based on free-text questions or narrative case descriptions. While this format facilitates interaction, it may not preserve the quantitative structure generated by upstream assessments. Information such as impairment grade, class-probability distributions, scale measurements, and other structured functional findings may be simplified or omitted when converted into short text queries. Therefore, the retrieved evidence may be relevant to general clinical problems but not perfectly aligned with an individual's measured functional status. This distinction is important in rehabilitation decision support, where recommendations may depend on specific findings from cognitive scales, SPPB, grip strength, TUG, or fall history. Therefore, establishing a clearer link between quantitative assessment and evidence retrieval is essential.

2.4 Personalized Rehabilitation Decision Support

Clinical decision support requires more than accurate prediction. In many healthcare applications, model outputs must ultimately be translated into recommendations or plans that are interpretable, evidence-informed, and suitable for professional review [34]. A predicted impairment grade can summarize overall severity, but it does not by itself identify which functions require attention, what evidence supports a

particular rehabilitation strategy, or which precautions should be considered.

Integrating data-driven models with domain knowledge provides a potential approach for addressing this challenge. Structured representations, clinical rules, retrieved evidence, and explicit constraints can complement machine learning outputs when decisions require more than classification or risk estimation [35]. This is particularly relevant to rehabilitation, where recommendations depend on both measured functional status and evidence derived from clinical guidelines and expert knowledge.

Existing clinical recommendation systems have combined machine learning with expert rules or guideline information, although many remain focused on risk prediction or categorical recommendations [36]. LLMs provide greater flexibility in producing personalized and readable content, but generation based only on free-form input can make the relationship between quantitative findings, supporting evidence, and final recommendations difficult to inspect or reproduce [37]. Therefore, an important challenge is to establish a transparent link between structured assessment information, external clinical knowledge, and personalized rehabilitation decision support.

The framework developed in this study addresses this challenge through a structured assessment-to-decision interface. Cognitive and motor assessment results are retained together with original measurements, deterministic scale summaries, demographic information, and attention-derived relevance cues to construct a structured functional profile. This profile conditions domain-specific evidence retrieval and subsequent rehabilitation recommendation generation under predefined safety constraints. In this way, quantitative assessment information is preserved throughout the transition from functional grading to evidence-informed rehabilitation decision support rather than being reduced to a single prediction label.

3. Proposed Method

3.1 System Overview

This framework organizes cognitive-motor assessment and rehabilitation decision support into a continuous quantitative decision support pathway. First, heterogeneous demographic, cognitive, and motor measurement data are characterized based on clinical domains and processed by BDCMT to assess cognitive and motor impairment status. The results are not presented as final predictive outputs but are organized into a structured functional profile containing predicted grades, class-probability distributions, original measurements, deterministic scale summaries, and attention-derived relevance cues. This profile provides a clear interface between quantitative assessment and rehabilitation decision support and guides evidence retrieval and subsequent recommendation generation.

For each participant, the input contains a demographic feature vector $\mathbf{x}_{\text{demo}}$, a cognitive feature vector $\mathbf{x}_{\text{cog}}$, and a motor feature vector $\mathbf{x}_{\text{mot}}$:

$$\mathbf{x}_{\text{demo}} \in \mathbb{R}^{n_{\text{demo}}}, \qquad \mathbf{x}_{\text{cog}} \in \mathbb{R}^{n_{\text{cog}}}, \qquad \mathbf{x}_{\text{mot}} \in \mathbb{R}^{n_{\text{mot}}} \tag{1}$$

where n_{demo}, n_{cog}, n_{mot} denote the numbers of demographic, cognitive, and motor variables, respectively.

The input features contain item-level measurements from cognitive and motor assessments, including

MoCA-B, MMSE, SPPB, grip strength, TUG, and fall history. Demographic variables include age, sex, and education level.

The corresponding cognitive and motor impairment labels are denoted by:

$$y_{\text{cog}} \in \{0,1,\dots,K_{\text{cog}}-1\}, \qquad y_{\text{mot}} \in \{0,1,\dots,K_{\text{mot}}-1\} \tag{2}$$

where K_{cog} and K_{mot} denote the numbers of cognitive and motor severity classes. Both assessment tasks contain four ordered severity levels: normal, mild, moderate, and severe impairment.

BDCMT predicts the cognitive and motor grades while retaining the corresponding class-probability distributions and attention information. Deterministic scoring rules are additionally applied to the original assessment measurements to obtain clinically interpretable scale summaries. These quantitative results form the structured functional profile used throughout the subsequent knowledge-grounding stage. The profile is used to construct domain-specific retrieval queries, while the retrieved evidence, quantitative assessment context, and safety constraints jointly condition QC-RAG-LLM during personalized rehabilitation recommendation generation. The final plan is provided for clinician review. **Fig. 1** illustrates the overall framework, and **Algorithm 1** summarizes its workflow.

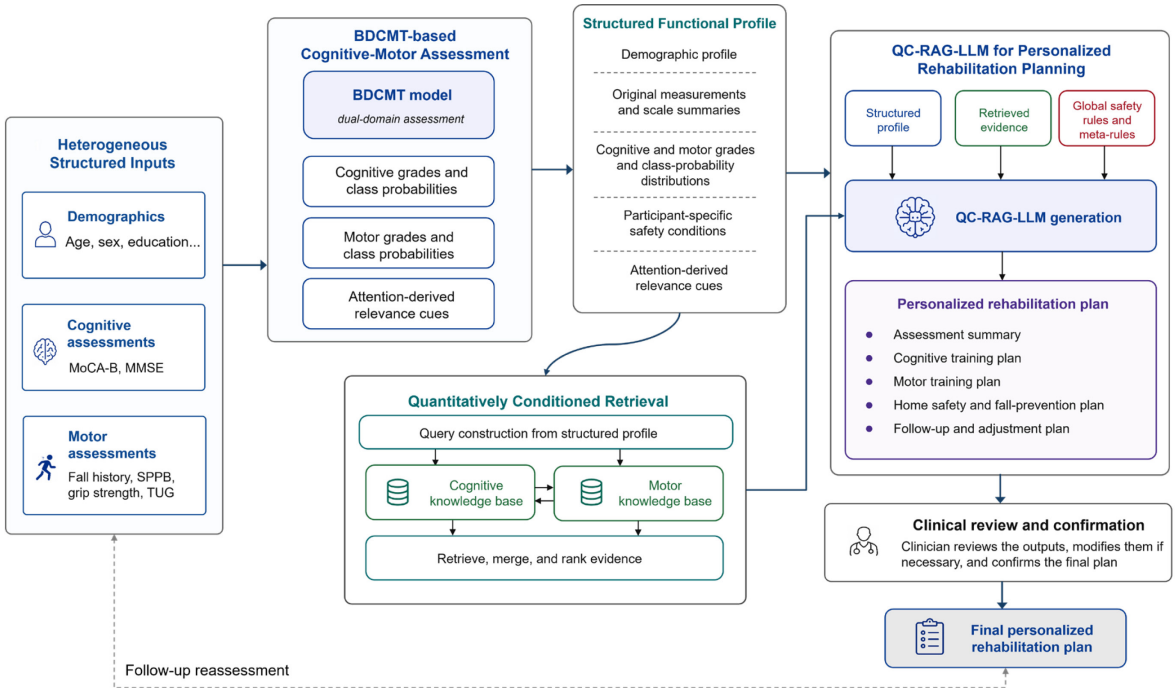


Fig. 1 Overall framework of the proposed system for cognitive-motor assessment and personalized rehabilitation decision support.

Algorithm 1. End-to-end cognitive-motor assessment and rehabilitation decision support

Input: Demographic features $\mathbf{x}_{\text{demo}}$, cognitive features $\mathbf{x}_{\text{cog}}$, motor features $\mathbf{x}_{\text{mot}}$, trained BDCMT parameters θ, cognitive knowledge base KB_{cog}, motor knowledge base KB_{mot}, embedding model $E(\cdot)$, deployed LLM P_{Θ}, number of selected features k, retrieval number K.

Output: Cognitive grade $\hat{y}_{\text{cog}}$, motor grade $\hat{y}_{\text{mot}}$, structured functional profile $\mathcal{S}$ and personalized rehabilitation recommendation y^{*}.

Offline preparation

1. Collect and preprocess cognitive and motor rehabilitation documents.
2. Split documents into text chunks.
3. Construct vector indices I_{cog} and I_{mot} using the embedding model $E(\cdot)$.
4. Load the trained BDCMT parameters θ.

Online assessment and recommendation generation

5. Load $\mathbf{x}_{\text{demo}}$, $\mathbf{x}_{\text{cog}}$ and $\mathbf{x}_{\text{mot}}$.
6. Apply the predefined preprocessing pipeline.
7. Generate demographic, cognitive, and motor feature tokens.
8. Encode cognitive and motor streams using the dual-stream Transformer encoders.
9. Apply reliability-gated bidirectional cross-attention between the two streams.
10. Obtain class probability distributions $\mathbf{p}_{\text{cog}}$ and $\mathbf{p}_{\text{mot}}$.
11. Compute predicted grades: $\hat{y}_{\text{cog}} = \arg\max_{j} p_{\text{cog},j}$, $\hat{y}_{\text{mot}} = \arg\max_{j} p_{\text{mot},j}$.
12. Extract attention-derived relevance scores and select the top k high-relevance features.
13. Construct the structured functional profile $\mathcal{S}$ from predicted grades, probabilities, original measurements, scale summaries, and relevance cues.
14. Construct retrieval queries from $\mathcal{S}$.
15. Retrieve the top K chunks from I_{cog} and I_{mot}.
16. Combine retrieved chunks as evidence context $\mathcal{C}_{\text{ret}}$.
17. Construct the final prompt using $\mathcal{S}$, $\mathcal{C}_{\text{ret}}$, and safety instructions $\mathcal{R}_{\text{meta}}$.
18. Generate the rehabilitation recommendation y^{*} using P_{Θ}.
19. Return $\hat{y}_{\text{cog}}$, $\hat{y}_{\text{mot}}$, $\mathcal{S}$ and y^{*}.

3.2 Reliability-Gated Bidirectional Domain-Aware Cross-Attention Medical Transformer

3.2.1 Feature-Specific Domain-Aware Tokenization

Clinical tabular data do not form a natural sequence. Each column has a predefined clinical meaning, and the column order is determined by the assessment form rather than temporal or linguistic order. Therefore, BDCMT does not use conventional sequential positional encoding. Instead, each feature is mapped into a token through a feature-specific learnable projection.

Let x_i^r denote the i-th scalar feature in data domain $r \in \{\text{demo}, \text{cog}, \text{mot}\}$. The corresponding feature token is calculated as:

$$\mathbf{v}_i^r = x_i^r \mathbf{w}_i^r + \mathbf{b}_i^r \tag{3}$$

where $\mathbf{w}_i^r \in \mathbb{R}^d$ and $\mathbf{b}_i^r \in \mathbb{R}^d$ are the feature-specific projection weight and bias. The embedding

dimension is denoted by d.

To distinguish the clinical source of each feature, a learnable domain embedding $\mathbf{e}^r \in R^d$ is added to the feature token:

$$\mathbf{h}_i^r = \text{Dropout}(\mathbf{v}_i^r + \mathbf{e}^r) \tag{4}$$

This design allows each variable to be identified by its own projection parameters while also preserving its domain information. Demographic tokens are shared by both the cognitive and motor streams. Learnable classification tokens are added to the two streams. The initial token sequences are defined as:

$$\begin{cases} \mathbf{H}_{\text{cog}}^0 = [\mathbf{h}_{\text{CLS}}^{\text{cog}}; \mathbf{H}_{\text{demo}}; \mathbf{H}_{\text{cog}}] \\ \mathbf{H}_{\text{mot}}^0 = [\mathbf{h}_{\text{CLS}}^{\text{mot}}; \mathbf{H}_{\text{demo}}; \mathbf{H}_{\text{mot}}] \end{cases} \tag{5}$$

where $\mathbf{H}_{\text{demo}}$, $\mathbf{H}_{\text{cog}}$, $\mathbf{H}_{\text{mot}}$ denote the demographic, cognitive, and motor token matrices, respectively. The semicolon denotes concatenation along the token dimension.

3.2.2 Dual-Stream Self-Attention Encoding

The cognitive and motor token sequences are encoded by two independent Transformer encoders. This design preserves the domain-specific structure of the two assessment streams before cross-domain interaction. Each encoder consists of multi-head self-attention, feed-forward networks, layer normalization, residual connections, and dropout.

Let $l \in \{1, \dots L\}$ denote the encoder-layer index, where L is the total number of encoder layers. For stream $r \in \{\text{cog}, \text{mot}\}$, the l-th encoder layer is expressed as:

$$\widetilde{\mathbf{H}}_r^l = \mathbf{H}_r^{l-1} + \text{MSA}\left(\text{LN}(\mathbf{H}_r^{l-1})\right) \tag{6}$$

$$\mathbf{H}_r^l = \widetilde{\mathbf{H}}_r^l + \text{FFN}\left(\text{LN}\left(\widetilde{\mathbf{H}}_r^l\right)\right) \tag{7}$$

where $\text{MSA}(\cdot)$ denotes multi-head self-attention, $\text{FFN}(\cdot)$ denotes the feed-forward network and $\text{LN}(\cdot)$ denotes layer normalization. The pre-normalization structure improves training stability and preserves the original representation through residual connections [38].

After the final encoder layer, the cognitive and motor representations are denoted by $\mathbf{H}_{\text{cog}}$ and $\mathbf{H}_{\text{mot}}$, respectively. The two streams first learn within-domain feature interactions. Cross-domain information is then exchanged through the bidirectional cross-attention module.

3.2.3 Reliability-Gated Bidirectional Cross-Attention

To model interactions between the cognitive and motor domains, BDCMT applies bidirectional cross-attention between the two encoding streams. Unlike direct cross-attention residual updates, the proposed module introduces a sample-level reliability gating to modulate the amount of cross-domain context injected into each stream. This design allows for information exchange between cognitive and motor representations while reducing the impact of unreliable cross-domain cues.

(1) Motor Querying Cognitive Representations

The query, key, and value matrices are defined as:

$$\mathbf{Q}_{\mathrm{mot}} = \mathbf{H}_{\mathrm{mot}}\mathbf{W}_Q^m, \qquad \mathbf{K}_{\mathrm{cog}} = \mathbf{H}_{\mathrm{cog}}\mathbf{W}_K^c, \qquad \mathbf{V}_{\mathrm{cog}} = \mathbf{H}_{\mathrm{cog}}\mathbf{W}_V^c \tag{8}$$

where $\mathbf{W}_Q^m, \mathbf{W}_K^c, \mathbf{W}_V^c$ are learnable projection matrices. Superscripts *c* and *m* indicate parameters associated with the cognitive and motor streams, respectively.

The cognitive context retrieved by the motor stream is:

$$\mathbf{C}_{\mathrm{cog}\to\mathrm{mot}} = \mathrm{MHA}\left(\mathbf{Q}_{\mathrm{mot}}, \mathbf{K}_{\mathrm{cog}}, \mathbf{V}_{\mathrm{cog}}\right) \tag{9}$$

where $\mathrm{MHA}(\cdot)$ denotes standard multi-head attention, including head-wise scaled dot-product attention, concatenation, and output projection. To regulate the contribution of cognitive context to the motor stream, a reliability gate is computed from the classification-token representations:

$$\mathbf{g}_{\mathrm{cog}\to\mathrm{mot}} = \sigma\left(\mathrm{MLP}_{\mathrm{g}}^{\mathrm{m}}\left(\left[\mathbf{h}_{\mathrm{cog}}^{\mathrm{CLS}}; \mathbf{h}_{\mathrm{mot}}^{\mathrm{CLS}}; \left|\mathbf{h}_{\mathrm{cog}}^{\mathrm{CLS}} - \mathbf{h}_{\mathrm{mot}}^{\mathrm{CLS}}\right|\right]\right)\right) \tag{10}$$

where $\sigma(\cdot)$ denotes the sigmoid function, and $\mathrm{MLP}_{\mathrm{g}}^{\mathrm{m}}(\cdot)$ denotes the gate network for cognitive-to-motor information modulation.

The gate output is a d-dimensional sample-level vector and is shared across token positions. The enhanced motor representation is:

$$\widehat{\mathbf{H}}_{\mathrm{mot}} = \mathrm{LN}\left(\mathbf{H}_{\mathrm{mot}} + \mathrm{Dropout}\left(\mathbf{g}_{\mathrm{cog}\to\mathrm{mot}} \odot \mathbf{C}_{\mathrm{cog}\to\mathrm{mot}}\right)\right) \tag{11}$$

where $\odot$ denotes element-wise multiplication with broadcasting along the token dimension. This branch captures cognitive information that contributes to motor impairment assessment, while the gate reduces indiscriminate cross-domain information injection.

(2) Cognitive Querying Motor Representations

The reverse branch is defined symmetrically. Cognitive features provide the query, while motor features provide the key and value:

$$\mathbf{Q}_{\mathrm{cog}} = \mathbf{H}_{\mathrm{cog}}\mathbf{W}_Q^c, \qquad \mathbf{K}_{\mathrm{mot}} = \mathbf{H}_{\mathrm{mot}}\mathbf{W}_K^m, \qquad \mathbf{V}_{\mathrm{mot}} = \mathbf{H}_{\mathrm{mot}}\mathbf{W}_V^m \tag{12}$$

The motor context retrieved by the cognitive stream is:

$$\mathbf{C}_{\mathrm{mot}\to\mathrm{cog}} = \mathrm{MHA}\left(\mathbf{Q}_{\mathrm{cog}}, \mathbf{K}_{\mathrm{mot}}, \mathbf{V}_{\mathrm{mot}}\right) \tag{13}$$

Similarly, the reliability gate for motor-to-cognitive information modulation is computed as:

$$\mathbf{g}_{\mathrm{mot}\to\mathrm{cog}} = \sigma\left(\mathrm{MLP}_{\mathrm{g}}^{\mathrm{c}}\left(\left[\mathbf{h}_{\mathrm{cog}}^{\mathrm{CLS}}; \mathbf{h}_{\mathrm{mot}}^{\mathrm{CLS}}; \left|\mathbf{h}_{\mathrm{cog}}^{\mathrm{CLS}} - \mathbf{h}_{\mathrm{mot}}^{\mathrm{CLS}}\right|\right]\right)\right) \tag{14}$$

The enhanced cognitive representation is:

$$\widehat{\mathbf{H}}_{\mathrm{cog}} = \mathrm{LN}\left(\mathbf{H}_{\mathrm{cog}} + \mathrm{Dropout}\left(\mathbf{g}_{\mathrm{mot}\to\mathrm{cog}} \odot \mathbf{C}_{\mathrm{mot}\to\mathrm{cog}}\right)\right) \tag{15}$$

Through these two branches, BDCMT models complementary interactions between cognitive and motor representations. The reliability gates are used as model-level modulation signals for cross-domain context injection. The attention matrices are retained during inference to derive feature-level relevance scores, which are treated as model-level relevance cues rather than causal explanations.

3.2.4 Cognitive and Motor Prediction

The classification tokens from the enhanced cognitive and motor representations are passed to two task-specific classifiers. The cognitive and motor logit vectors are:

$$\mathbf{o}_{\mathrm{cog}} = \mathrm{g}_{\mathrm{cog}}\left(\hat{\mathbf{h}}_{\mathrm{cog}}^{\mathrm{CLS}}\right), \qquad \mathbf{o}_{\mathrm{mot}} = \mathrm{g}_{\mathrm{mot}}\left(\hat{\mathbf{h}}_{\mathrm{mot}}^{\mathrm{CLS}}\right) \tag{16}$$

where $\mathrm{g}_{\mathrm{cog}}(\cdot)$ and $\mathrm{g}_{\mathrm{mot}}(\cdot)$ denote the task-specific classification heads for cognitive and motor impairment grading. The vectors $\hat{\mathbf{h}}_{\mathrm{cog}}^{\mathrm{CLS}}$ and $\hat{\mathbf{h}}_{\mathrm{mot}}^{\mathrm{CLS}}$ are the classification-token vectors extracted from the enhanced cognitive and motor representations.

The class probability distributions are obtained by softmax normalization:

$$\mathbf{p}_{\mathrm{cog}} = \mathrm{softmax}\left(\mathbf{o}_{\mathrm{cog}}\right), \qquad \mathbf{p}_{\mathrm{mot}} = \mathrm{softmax}(\mathbf{o}_{\mathrm{mot}}) \tag{17}$$

Let j denote the class index. The predicted cognitive and motor grades are:

$$\hat{y}_{\mathrm{cog}} = \arg\max_{j} p_{\mathrm{cog},j}\,, \qquad \hat{y}_{\mathrm{mot}} = \arg\max_{j} p_{\mathrm{mot},j} \tag{18}$$

where $p_{\mathrm{cog},j}$ and $p_{\mathrm{mot},j}$ denote the predicted probabilities of class j for the cognitive and motor tasks.

The maximum predicted probabilities are:

$$s_{\mathrm{cog}} = \max_{j} p_{\mathrm{cog},j}\,, \qquad s_{\mathrm{mot}} = \max_{j} p_{\mathrm{mot},j} \tag{19}$$

The values s_{cog} and s_{mot} are displayed as model confidence scores. They are not assumed to be calibrated clinical probabilities. The complete architecture of BDCMT is summarized in **Fig. 2**.

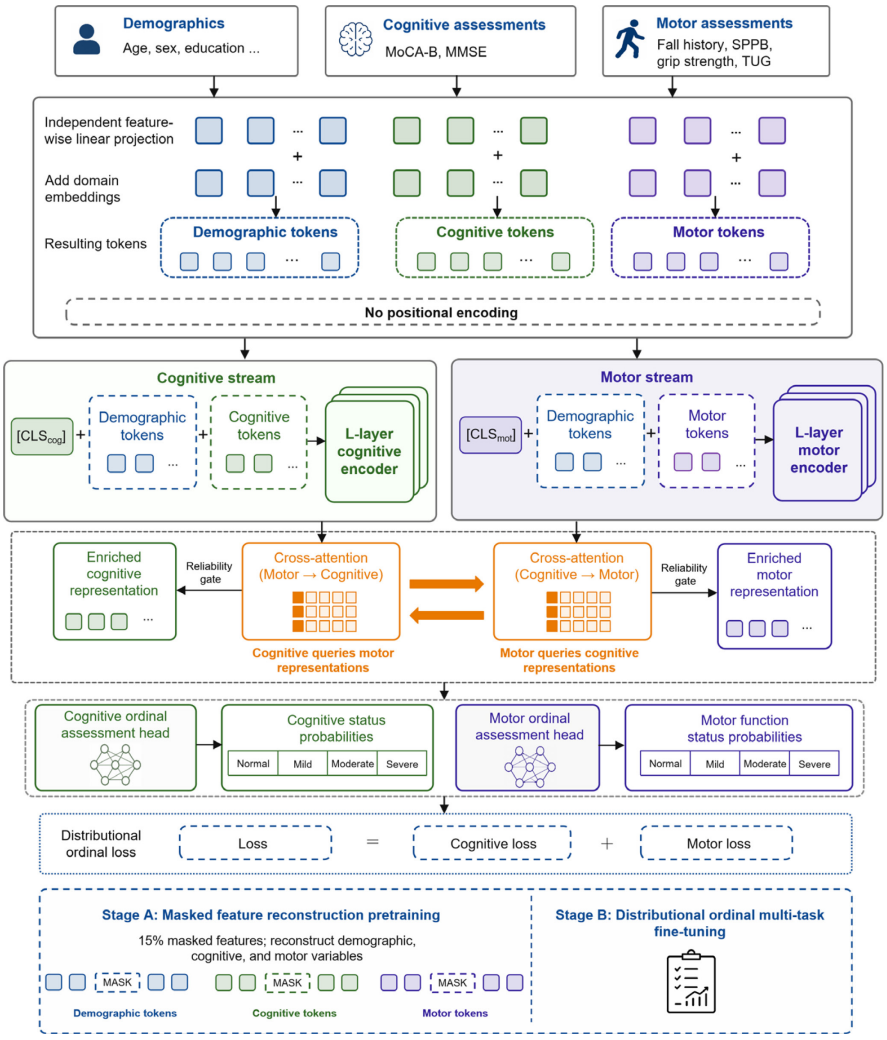


Fig. 2 Architecture of the BDCMT model for cognitive-motor assessment.

3.3 Two-Stage Optimization

3.3.1 Masked Feature Reconstruction

To improve representation learning from moderate-sized structured assessment data, BDCMT is first initialized using masked feature reconstruction. During this stage, a subset of demographic, cognitive, and motor variables is randomly masked. The model is trained to reconstruct the original standardized values of the masked variables.

Let $\mathcal{M}$ denote the set of masked feature indices. The reconstruction loss is defined as:

$$\mathcal{L}_{\text{pre}} = \frac{1}{|\mathcal{M}|} \sum_{i \in \mathcal{M}} (\hat{x}_i - x_i)^2 \tag{20}$$

where x_i is the original standardized clinical value, and $\hat{x}_i$ is the reconstructed value. Only masked variables contribute to the pretraining objective.

Masked feature reconstruction was performed using only the training partition of each experimental split. Standardization parameters were estimated from the training set only, and neither validation nor test samples were used for reconstruction-based pretraining or parameter initialization. After pretraining, the reconstruction heads were no longer used, and the pretrained BDCMT parameters were transferred to supervised fine-tuning.

3.3.2 Distributional Ordinal Loss

Cognitive and motor impairment grades are ordered. Standard cross-entropy loss can distinguish classes, but it does not directly account for the distance between severity levels. For example, predicting severe impairment as normal should receive a larger penalty than predicting it as moderate impairment. To better reflect this ordinal structure, a distributional ordinal loss is introduced.

For a task with C classes, let $\mathbf{y} = [y_0, y_1, \ldots, y_{C-1}]$ denote the one-hot encoded label vector, and let $\mathrm{y}_{\text{true}} \in \{0,1, \ldots, C-1\}$ denote the true ordinal grade. The predicted class probability vector is $\mathbf{p} = [p_0, p_1, \ldots, p_{C-1}]$. The cross-entropy loss is:

$$\mathcal{L}_{\text{ce}} = -\sum_{j=0}^{C-1} y_j \log(p_j) \tag{21}$$

To encode the proximity between severity grades, a distance-aware soft target distribution is constructed as:

$$\boldsymbol{q}_j = \frac{\exp\left(-\frac{|j - y_{\text{true}}|}{\tau}\right)}{\sum_{r=0}^{C-1} \exp\left(-\frac{|r - y_{\text{true}}|}{\tau}\right)}, j = 0,1, \ldots, C-1 \tag{22}$$

where τ controls the smoothness of the target distribution. The value of τ is 1. Classes closer to the true grade receive larger target probabilities, whereas distant classes receive smaller probabilities.

The soft ordinal classification loss is defined as:

$$\mathcal{L}_{\text{soft}} = -\sum_{j=0}^{C-1} \boldsymbol{q}_j \log(p_j) \tag{23}$$

To further constrain the ordered structure of the probability distribution, the cumulative distributions of p and q are computed as:

$$F_j(p) = \sum_{r=0}^{j} p_r\,, \qquad F_j(q) = \sum_{r=0}^{j} q_r \tag{24}$$

The cumulative distribution discrepancy is defined as:

$$\mathcal{L}_{\text{cum}} = \frac{1}{C-1}\sum_{j=0}^{C-2} \text{SmoothL1}\left(F_j(p) - F_j(q)\right) \tag{25}$$

The distributional ordinal loss is then:

$$\mathcal{L}_{\text{dol}} = \mathcal{L}_{\text{soft}} + \beta\mathcal{L}_{\text{cum}} \tag{26}$$

where β controls the contribution of the cumulative distribution constraint and is set to 0.5.

For cognitive and motor grading, the task-specific losses are:

$$\begin{cases} \mathcal{L}_{\text{cog}} = \mathcal{L}_{\text{ce}}^{\text{cog}} + \lambda_{\text{dol}}\mathcal{L}_{\text{dol}}^{\text{cog}} \\ \mathcal{L}_{\text{mot}} = \mathcal{L}_{\text{ce}}^{\text{mot}} + \lambda_{\text{dol}}\mathcal{L}_{\text{dol}}^{\text{mot}} \end{cases} \tag{27}$$

where λ_{dol} is the ordinal regularization coefficient and is set to 0.2.

The total multi-task loss is:

$$\mathcal{L}_{\text{total}} = \omega_{\text{cog}}\mathcal{L}_{\text{cog}} + \omega_{\text{mot}}\mathcal{L}_{\text{mot}} \tag{28}$$

where ω_{cog} and ω_{mot} are the task weights. In this study, cognitive and motor tasks are assigned equal importance, and $\omega_{\text{cog}} = \omega_{\text{mot}} = 0.5$.

3.4 Structured Assessment-to-Decision Interface

After BDCMT inference, the assessment results are converted into a structured functional profile. This profile bridges quantitative assessment and personalized rehabilitation recommendation generation and contains demographic information, predicted grades, class-probability distributions, original clinical measurements, deterministic scale summaries, attention-derived relevance cues, and safety constraints.

Attention-derived relevance cues are computed from the final self-attention layer and the two cross-attention branches. Attention weights are averaged across heads, and the classification-token query row is retained. Four relevance categories are constructed: cognitive within-domain relevance, motor within-domain relevance, motor features used during cognitive assessment, and cognitive features used during motor assessment. The top k features are selected independently in each category:

$$\mathcal{T}_\ell = \text{TopK}_k\left(\{(x_i, r_i^\ell)\}_{i\in\Omega_\ell}\right), \qquad \ell \in \{\text{cog-self}, \text{mot-self}, \text{mot} \to \text{cog}, \text{cog} \to \text{mot}\} \tag{29}$$

For each selected feature, the system records its feature name, original measurement, and normalized relevance weight. The structured relevance set is:

$$\mathcal{S}_{\text{key}} = \cup_\ell \left\{\left(x_i, v_{i,\text{raw}}, \tilde{r}_i^\ell\right) \mid i \in \mathcal{T}_\ell\right\} \tag{30}$$

The complete structured functional profile is then defined as：

$$\mathcal{S} = \{\mathbf{x}_{\text{demo}}, \hat{y}_{\text{cog}}, \hat{y}_{\text{mot}}, \mathbf{p}_{\text{cog}}, \mathbf{p}_{\text{mot}}, \mathcal{S}_{\text{meas}}, \mathcal{S}_{\text{key}}, \mathcal{S}_{\text{scale}}, \mathcal{S}_{\text{safe}}\} \quad (31)$$

where $\mathcal{S}_{\text{meas}}$ denotes selected original assessment measurements. $\mathcal{S}_{\text{scale}}$ denotes deterministic summaries of MoCA-B, MMSE, SPPB, grip strength, TUG, and fall history, and $\mathcal{S}_{\text{safe}}$ denotes safety constraints for personalized rehabilitation recommendation generation.

These cues are described as high-relevance indicators rather than abnormal findings or causal factors. Their clinical meaning should be interpreted together with the original measurements and scale summaries.

The structured functional profile is stored in JavaScript Object Notation (JSON) format and retained throughout retrieval and generation. It therefore serves as a common quantitative representation of the participant's assessed functional state, while also allowing the intermediate information used for rehabilitation decision support to be inspected before the final plan is generated.

3.5 Quantitatively Conditioned Retrieval-Augmented Generation

QC-RAG-LLM performs assessment-based evidence retrieval and rehabilitation recommendation generation through a structured assessment-to-decision interface. The structured functional profile is used both to construct domain-specific retrieval queries and to define generation constraints, ensuring that quantitative assessment information remains explicit throughout the knowledge selection and recommendation generation process. Cognitive and motor rehabilitation evidence is retrieved separately and then combined with the structured profile and safety constraints. **Fig. 3** illustrates this process.

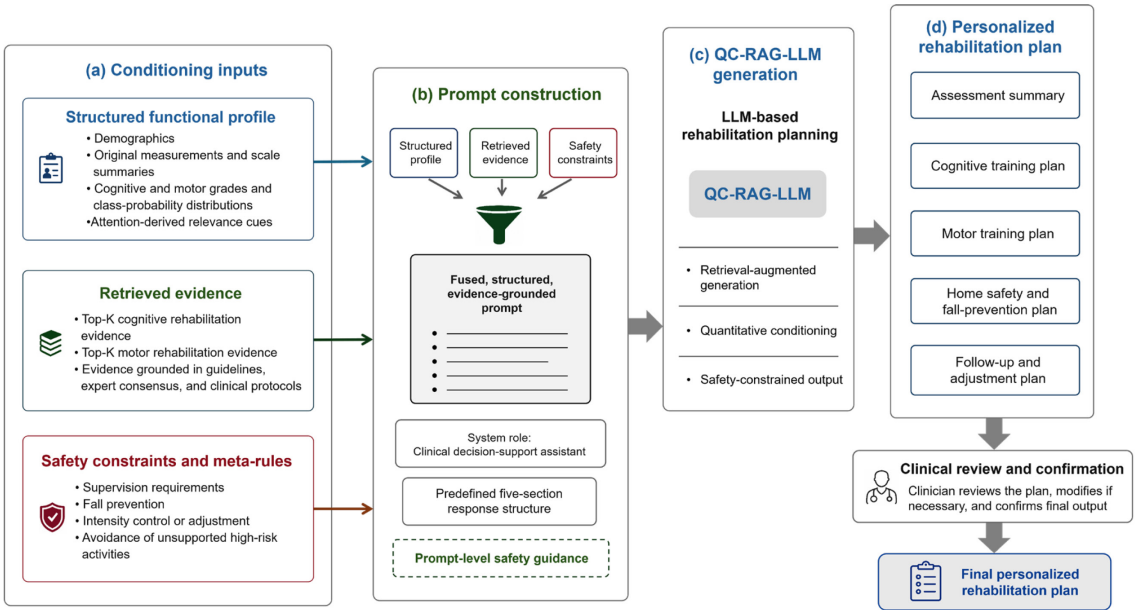


Fig. 3 Assessment-conditioned prompt construction and personalized rehabilitation recommendation generation in QC-RAG-LLM.

3.5.1 Domain-Specific Knowledge Bases

The QC-RAG-LLM uses two separate knowledge bases: a cognitive knowledge base KB_{cog} and a motor knowledge base KB_{mot}. The initial corpus comprised more than 50 clinical guidelines, expert consensus documents, and rehabilitation-related publications. After clinician review, 35 documents

considered relevant and practical for cognitive-motor assessment and rehabilitation were retained and assigned to the corresponding domain-specific collection.

The PDF documents are parsed and divided into text chunks. Text embeddings are generated using the embedding model, and separate cognitive and motor vector indices are constructed and stored for subsequent retrieval. Maintaining separate indices avoids direct competition between cognitive and motor documents and ensures that both domains contribute evidence to the generation context.

Let $\mathcal{D}_{\mathrm{cog}}$ and $\mathcal{D}_{\mathrm{mot}}$ denote the document collections for cognitive and motor rehabilitation. For domain $s \in \{\mathrm{cog},\ \mathrm{mot}\}$, each document is divided into text chunks. The embedding model $E(\cdot)$ maps each chunk into a dense vector. The domain-specific vector index is:

$$I_s = \{E(c_{i,d}) \mid d \in \mathcal{D}_s, \qquad i = 1, \ldots, N_d\} \tag{32}$$

where $c_{i,d}$ represents the *i*-th text chunk of document d. N_d is the number of chunks in that document.

3.5.2 Parallel Retrieval

Separate queries q_{cog} and q_{mot} are constructed from the structured functional profile $\mathcal{S}$. The cognitive query includes the predicted cognitive grade, reduced MMSE and MoCA-B domains, and cognitive rehabilitation keywords derived from the selected relevance cues. The motor query includes the predicted motor grade, grip-strength status, TUG classification, fall-history information, and predefined motor rehabilitation and safety keywords. Class-probability distributions remain part of the structured prompt context but are not directly inserted into the retrieval queries.

For domain $s \in$ {cog, mot}, the corresponding query is embedded as $E(q_s)$. The similarity between the query and a document chunk is measured using cosine similarity:

$$\mathrm{sim}\left(E(q_s), E(c_{i,d})\right) = \frac{E(q_s)^{\top} E(c_{i,d})}{||E(q_s)||_2 ||E(c_{i,d})||_2} \tag{33}$$

The retrieval engine searches I_{cog} and I_{mot} independently. The top *K* chunks are selected from each index. The combined retrieval context is:

$$\mathcal{C}_{\mathrm{ret}} = \mathrm{TopK}\left(E(q_{cog}), I_{\mathrm{cog}}\right) \oplus \mathrm{TopK}(E(q_{mot}), I_{\mathrm{mot}}) \tag{34}$$

where $\oplus$ denotes ordered concatenation. In the experiments, the top three chunks were retrieved from each domain-specific knowledge base. The retrieved chunks were merged into the evidence context, and each chunk retained its source filename and similarity score. No additional reranking model or fixed similarity threshold was applied.

By retrieving from the two indices independently, the strategy preserves separate cognitive and motor evidence channels and prevents either domain from being excluded solely because of cross-domain similarity ranking.

3.5.3 Quantitative Conditioning of the Large Language Model

The LLM receives the structured functional profile $\mathcal{S}$ and the retrieved evidence $\mathcal{C}_{\mathrm{ret}}$. The prompt also contains clinical safety and formatting instructions, denoted by $\mathcal{R}_{\mathrm{meta}}$. The complete prompt context is:

$$\mathcal{P} = \{\mathcal{S}, \mathcal{C}_{\text{ret}}, \mathcal{R}_{\text{meta}}\} \tag{35}$$

The generated rehabilitation recommendation is expressed as:

$$y^* = \arg\max_{y} P_{\Theta}(y \mid \mathcal{P}) \tag{36}$$

where y^* denote the intervention-related Sections 2 – 5 generated by the LLM and Θ denotes the fixed parameters of the deployed LLM. Quantitative conditioning is implemented through structured prompting rather than parameter fine-tuning.

The safety and formatting instructions $\mathcal{R}_{meta}$ guide the generation process. $\mathcal{S}_{\text{safe}}$ contains participant-specific safety conditions, whereas $\mathcal{R}_{meta}$ contains global generation and formatting rules. They require the plan to consider age, sex, education level, predicted grades, class probabilities, original measurements, deterministic scale summaries, and retrieved evidence. They also restrict unsafe recommendations. For participants with moderate or severe motor impairment, the prompt emphasizes supervision, stable support, fall prevention, and avoidance of unsupported high-risk activities. Attention-derived relevance cues are used to identify areas requiring additional consideration, but they are not described as causal factors.

3.5.4 Rehabilitation recommendation generation

The LLM generates the rehabilitation recommendation autoregressively:

$$P_{\Theta}(y \mid \mathcal{P}) = \prod_{t=1}^{T} P_{\Theta}(y_t \mid y_{<t}, \mathcal{P}) \tag{37}$$

where T denotes the output sequence length, y_t is the token generated at step t. $y_{<t}$ denotes all tokens generated before step t.

The final rehabilitation report contains five parts: (1) Comprehensive assessment and interpretation; (2) Cognitive rehabilitation training; (3) Motor rehabilitation training; (4) Home safety and fall prevention; (5) Follow-up assessment and plan adjustment.

The comprehensive assessment section is constructed directly from the structured functional profile to preserve consistency with the quantitative assessment results. QC-RAG-LLM is primarily responsible for generating the rehabilitation intervention sections under the retrieved evidence and predefined safety constraints. In this setting, the LLM organizes assessment-conditioned and evidence-grounded information rather than independently determining the participant's functional status. The generated recommendations are intended to support clinical decision-making and do not replace professional diagnosis or treatment.

3.6 Complexity Analysis

Let N_{cog} and N_{mot} denote the numbers of tokens in the cognitive and motor streams, d the token embedding dimension, L the number of encoder layers, M the total number of text chunks in the two knowledge bases, d_r the retrieval embedding dimension, and T the output length of the QC-RAG-LLM. For one participant, the main computational cost of BDCMT inference arises from dual-stream self-attention and

bidirectional cross-attention. The dominant attention-related time complexity is: $O\left(L\left(N_{\mathrm{cog}}^2 + N_{\mathrm{mot}}^2\right)d + 2N_{\mathrm{cog}}N_{\mathrm{mot}}d\right)$. The first term corresponds to self-attention in the cognitive and motor streams, and the second corresponds to cross-attention in both directions. The reliability gates involve linear projections and element-wise modulation and do not change the overall asymptotic complexity. Exact similarity search over the two knowledge bases requires $O(Md_r)$ time per query, while approximate nearest-neighbor indexing can reduce the practical retrieval cost. The generation cost of QC-RAG-LLM depends on the architecture of the deployed model, the prompt length, the output length *T*, and the decoding settings.

The space complexity of BDCMT inference is mainly determined by token representations and attention maps: $O\left(\left(N_{\mathrm{cog}} + N_{\mathrm{mot}}\right)d + N_{\mathrm{cog}}^2 + N_{\mathrm{mot}}^2 + 2N_{\mathrm{cog}}N_{\mathrm{mot}}\right)$. The retrieval indices require $O(Md_r)$ storage. Because QC-RAG-LLM is accessed as a fixed external service, its parameters are not included in the trainable parameter or memory complexity of the proposed framework.

4. Experiments and Results

4.1 Dataset and Experimental Setup

This study included 977 community-dwelling older adults with complete demographic, cognitive, and motor assessment records. Demographic variables included age, sex, and education level. Cognitive function was assessed using the Chinese version of the Montreal Cognitive Assessment Basic (MoCA-B) and the Chinese version of the Mini-Mental State Examination (MMSE) [39]. Motor function was evaluated using fall history, the short physical performance battery (SPPB), grip strength, and the timed up and go (TUG) test [40].

Cognitive and motor impairment grades were assigned by experienced physicians according to the scale scores, original measurements, and predefined grading criteria. Both tasks were categorized into four ordered levels: normal, mild, moderate, and severe impairment. These grades were used as reference labels for model training and evaluation.

The cognitive and motor assessment tables were matched at the participant level and divided into training, validation, and test sets with a ratio of 60%, 20%, and 20%, respectively. Stratified sampling was performed according to the joint cognitive-motor label distribution to maintain a similar severity distribution across different partitions. All preprocessing parameters, including feature standardization, were estimated using the training set only and then applied to the validation and test sets. For models involving self-supervised, contrastive, or reconstruction-based pretraining, only the training partition was used during pretraining. Hyperparameters were selected based on validation-set performance, while the test set was reserved for final evaluation.

The compared methods included TabM [19], GTD-MLP [41], FT-Transformer [42], MMoE [23], VIME [20], SCARF [21], TransTab [22], and the proposed BDCMT. Models were trained for 40 epochs with a batch size of 64 using AdamW, with a learning rate of 1×10^{-4} and a weight decay of 2×10^{-2}. Transformer-based models used comparable encoder configurations, including an embedding dimension of 128, three encoder

layers, four attention heads, a feed-forward dimension of 512, and a dropout rate of 0.1. Each experiment was repeated ten times using different random seeds. Within each run, all models were evaluated using the same participant-level split, preprocessing procedure, and validation-based model selection strategy. Across runs, the training, validation, and test partitions were regenerated using the same joint cognitive-motor label stratification procedure. The main configurations of all models are summarized in **Table 1**.

All deep learning experiments were conducted on a Windows 11 workstation equipped with an Intel Core i5-14490F CPU (2.80 GHz), 32 GB of RAM, and an NVIDIA GeForce RTX 5060 GPU with 8 GB of memory. QC-RAG-LLM was implemented using Qwen3.6-Plus through the DashScope service. The text-embedding-v2 model was used to construct the vector indices. During retrieval, the top three text chunks were retrieved independently from each domain-specific knowledge base. The maximum generation length was set to 1,500 tokens. The structured functional profile retained the selected attention-derived relevance cues for subsequent retrieval and personalized rehabilitation recommendation generation. All records were de-identified before external API processing.

Table 1 Main configurations of the comparison models and the proposed BDCMT.

Models	Parameter settings
TabM	BatchEnsemble-style parameter-efficient MLP; ensemble size K=32; two task-specific classification heads for cognitive and motor grading.
GTD-MLP	Shared MLP backbone with generalized triangular-distribution soft labels; left/right support parameters both set to 0.15; blending coefficient set to 1.0; two task-specific classification heads.
FT-Transformer	Single-stream feature-tokenized Transformer; shared CLS representation; PreNorm self-attention encoder with two task-specific classification heads.
MMoE	Multi-gate mixture-of-experts architecture; 4 shared experts; separate task-specific gates for cognitive and motor grading.
VIME	Mask prediction and feature reconstruction; corruption probability P_m set to 0.3; reconstruction weight set to 1.0; Gaussian noise standard deviation set to 0.008.
SCARF	Contrastive pre-training with random feature corruption and InfoNCE loss; corruption ratio set to 0.6; temperature parameter set to 0.1.
TransTab	Quantile-based column-wise tokenization using feature, value-bin, and separator tokens; number of bins set to 16; gated Transformer encoder.
BDCMT	Reliability-gated domain-aware medical Transformer with bidirectional cognitive-motor cross-attention; masked feature reconstruction pre-training with masking ratio set to 0.15; distributional ordinal loss with $\lambda_{\mathrm{dol}} = 0.2$, $\tau = 1$ and $\beta = 0.5$.

4.2 Evaluation of BDCMT

4.2.1 Comparison with Baseline Models

To evaluate the performance of BDCMT on cognitive and motor impairment grading tasks, this study selected seven representative baseline models: TabM, GTD-MLP, FT-Transformer, MMoE, VIME, SCARF, and TransTab. These methods cover different categories of tabular learning, including parameter-efficient modeling, ordinal learning, Transformer-based representation learning, multi-task learning, and self-supervised representation learning.

TabM was selected as a strong parameter-efficient ensemble-style MLP baseline for tabular data. GTD-MLP was included to represent ordinal classification methods using soft label distributions to capture the proximity between impairment levels. FT-Transformer and TransTab were used as representative Transformer-based tabular models that learn feature interactions through tokenization and attention mechanisms. MMoE was adopted as a multi-task learning baseline with shared experts and task-specific gating. VIME and SCARF were included to evaluate the effectiveness of self-supervised representation learning through feature reconstruction and contrastive learning, respectively. All models were trained and evaluated independently ten times using different random seeds, and the average performance was reported for comparison.

Table 2 shows the performance comparison results of each model in the cognitive and motor assessment grading task. For cognitive grading, BDCMT achieved the best mean performance across all evaluation metrics, including ACC, $\mathrm{AUC}_{\mathrm{OVR}}$, $\mathrm{F1(w)}$, MCC, and Grade MAE. Compared to the closest performing model, FT-Transformer, BDCMT improved its ACC from 0.8658 to 0.8867, MCC from 0.8082 to 0.8381, and Grade MAE from 0.1342 to 0.1133. BDCMT also showed the lowest SD across all five cognitive metrics, indicating good run-to-run stability. Among the comparison models, FT-Transformer remained the most competitive baseline, followed by TabM and TransTab on most cognitive metrics, whereas MMoE showed relatively weaker performance.

For motor grading, BDCMT also achieved the best mean across all evaluation metrics, with a more significant improvement over the baseline model. Compared to FT-Transformer, its ACC improved from 0.8592 to 0.9036, MCC from 0.8045 to 0.8658, and Grade MAE from 0.1413 to 0.0964. The improvement was more evident in the motor task, suggesting that cognitive-domain information can provide complementary cues for interpreting gait, balance, strength, and mobility decline. BDCMT also maintained lower SD values across the motor metrics, supporting its stability across repeated runs.

Overall, BDCMT achieved the strongest performance across the evaluated metrics and the lowest Grade MAE in both cognitive and motor impairment grading tasks. These results indicate that domain-aware dual-stream modeling, reliability-gated bidirectional cognitive-motor interaction, and distributional ordinal loss can jointly improve structured cognitive-motor assessment, with a more pronounced advantage in motor impairment grading.

Table 2 Performance comparison of different models on cognitive and motor impairment grading tasks.

Category	Metric	Statistic	TabM	GTD-MLP	FT-Transformer	MMoE	VIME	SCARF	TransTab	BDCMT
Cognitive Impairment	ACC↑	Mean	0.8571	0.7944	0.8658	0.7286	0.8051	0.7964	0.8474	**0.8867**
		SD	0.0243	0.0268	0.0223	0.0330	0.0175	0.0182	0.0193	**0.0157**
	AUC_{OVR}↑	Mean	0.9713	0.9530	0.9768	0.9413	0.9593	0.9561	0.9651	**0.9828**
		SD	0.0108	0.0108	0.0055	0.0200	0.0094	0.0093	0.0087	**0.0044**
	F1(w)↑	Mean	0.8556	0.7890	0.8645	0.6987	0.8010	0.7916	0.8463	**0.8863**
		SD	0.0252	0.0287	0.0231	0.0413	0.0192	0.0186	0.0197	**0.0160**
	MCC↑	Mean	0.7962	0.7050	0.8082	0.6260	0.7218	0.7069	0.7817	**0.8381**
		SD	0.0328	0.0375	0.0317	0.0439	0.0229	0.0271	0.0286	**0.0214**
	Grade MAE↓	Mean	0.1444	0.2082	0.1342	0.2735	0.1974	0.2056	0.1526	**0.1133**
		SD	0.0256	0.0258	0.0223	0.0368	0.0168	0.0188	0.0193	**0.0157**
Motor Impairment	ACC↑	Mean	0.8000	0.7668	0.8592	0.6888	0.7801	0.7730	0.8128	**0.9036**
		SD	0.0238	0.0207	0.0350	0.0386	0.0349	0.0254	0.0383	**0.0187**
	AUC_{OVR}↑	Mean	0.9534	0.9288	0.9721	0.8951	0.9334	0.9319	0.9568	**0.9830**
		SD	0.0115	0.0096	0.0076	0.0151	0.0119	0.0124	0.0104	**0.0038**
	F1(w)↑	Mean	0.7950	0.7509	0.8588	0.6241	0.7674	0.7602	0.8085	**0.9035**
		SD	0.0256	0.0284	0.0355	0.0473	0.0419	0.0315	0.0408	**0.0190**
	MCC↑	Mean	0.7198	0.6789	0.8045	0.5708	0.6956	0.6857	0.7385	**0.8658**
		SD	0.0357	0.0273	0.0470	0.0463	0.0442	0.0287	0.0513	**0.0262**
	Grade MAE↓	Mean	0.2082	0.2454	0.1413	0.4020	0.2485	0.2378	0.1913	**0.0964**
		SD	0.0289	0.0235	0.0356	0.0420	0.0523	0.0241	0.0400	**0.0187**

As shown in **Figs. 4** and **5**, the validation-set metrics of the comparison models gradually stabilized over the training epochs for both cognitive and motor impairment grading. BDCMT exhibited a faster convergence speed in both tasks, with its ACC, AUC_{OVR}, F1(w), and MCC improving rapidly in the early stages of training and maintaining high and relatively stable levels in subsequent stages. Grade MAE also decreased quickly and converged to the lowest range. FT-Transformer and TabM provided competitive baselines, whereas TransTab showed moderate performance. VIME and SCARF exhibited similar convergence patterns, while MMoE and GTD-MLP generally remained lower on most metrics. Overall, BDCMT showed good optimization efficiency and convergence stability in both tasks.

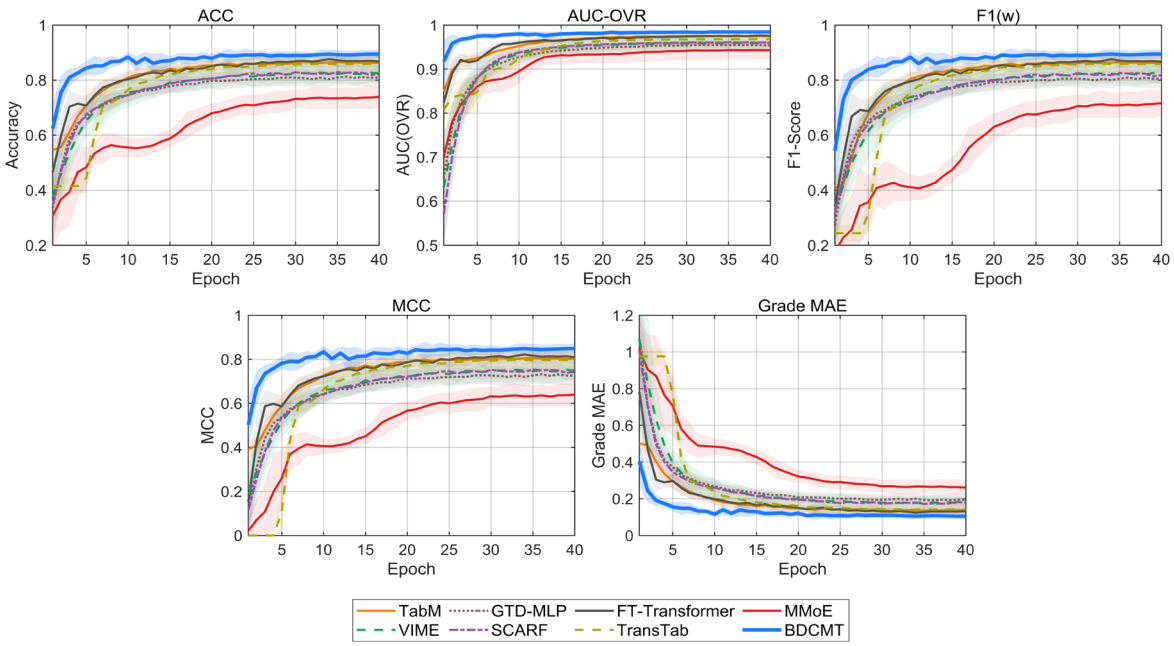


Fig. 4 Validation-set convergence curves of different models in cognitive function grading tasks.

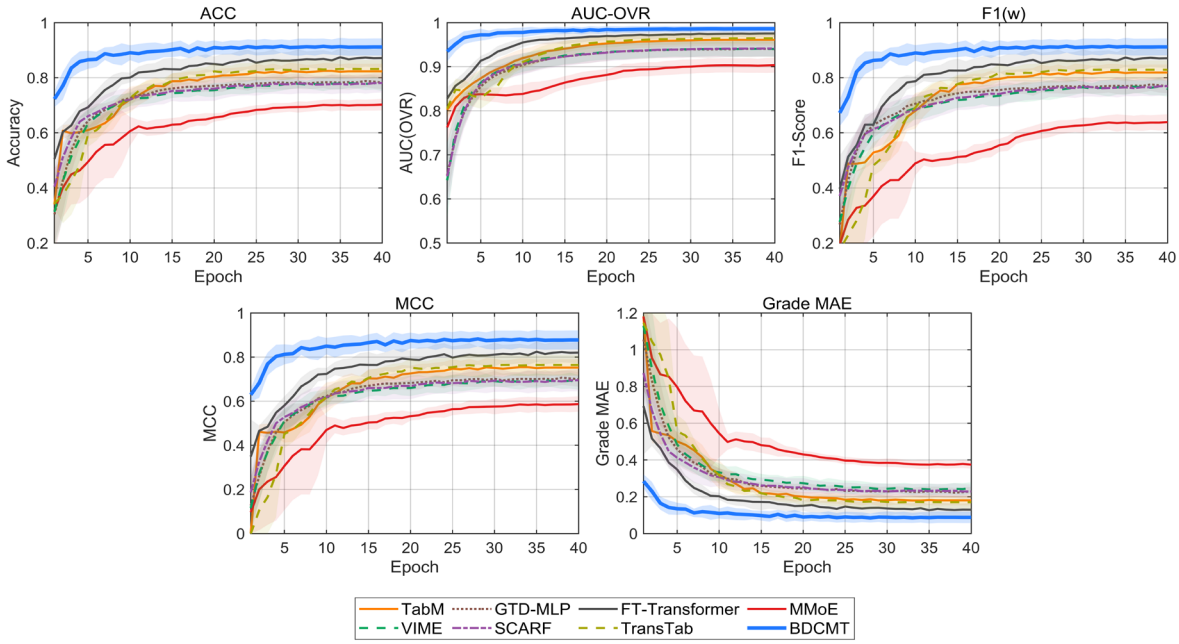


Fig. 5 Validation-set convergence curves of different models in the motor function grading tasks.

Fig. 6 shows the average performance heatmap of different models in the cognitive and motor impairment grading tasks. BDCMT achieved the highest values in ACC, AUC_{OVR}, F1(w), and MCC, and the lowest Grade MAE in both tasks. FT-Transformer was the strongest baseline overall, while TabM and TransTab also showed competitive results on several metrics. Compared with these baselines, BDCMT showed a clearer advantage in the motor grading task, suggesting that reliability-gated bidirectional cross-domain interaction is beneficial for modeling cognitive-motor functional dependence.

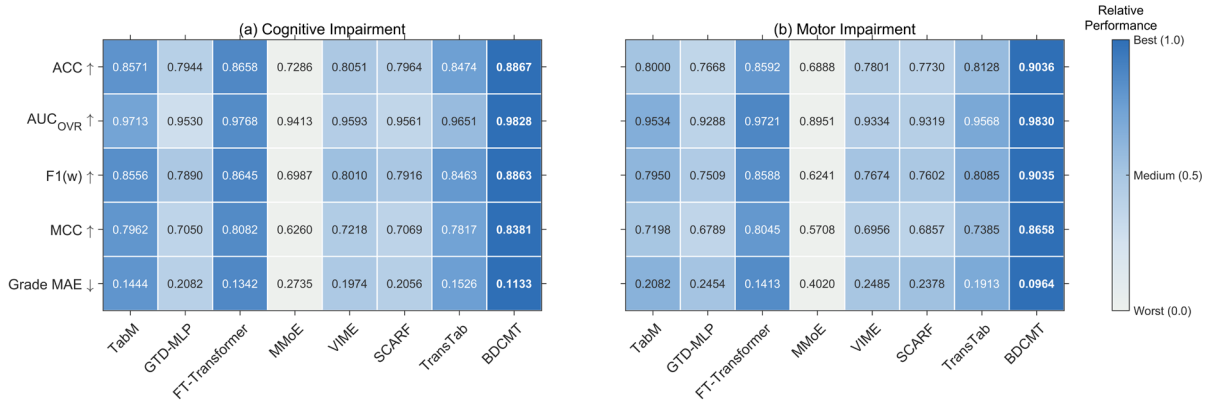


Fig. 6 Average performance heatmap of different models in cognitive and motor function grading tasks.

The confusion matrices for the cognitive and motor impairment grading tasks are shown in **Figs. 7** and **8**, respectively. In cognitive grading, BDCMT showed a clearer diagonal structure than the comparison models. Misclassifications mainly occurred between adjacent grades, such as normal and mild, or mild and moderate impairment. No obvious long-distance confusion was observed between normal and moderate/severe grades. FT-Transformer and TabM also showed relatively clear diagonal patterns, whereas GTD-MLP, VIME, SCARF, TransTab, and MMoE exhibited more confusion in the mild and moderate grades.

In motor grading, BDCMT again produced a strong diagonal distribution. Errors were mainly concentrated in adjacent levels, especially between mild and moderate or between moderate and severe impairment. FT-Transformer remained competitive, but it showed more adjacent misclassifications than BDCMT. TabM and TransTab achieved moderate performance, whereas GTD-MLP, VIME, SCARF, and MMoE showed more evident confusion between neighboring grades. Overall, BDCMT demonstrated fewer cross-level misclassifications in both tasks, indicating better consistency with the ordered structure of impairment severity.

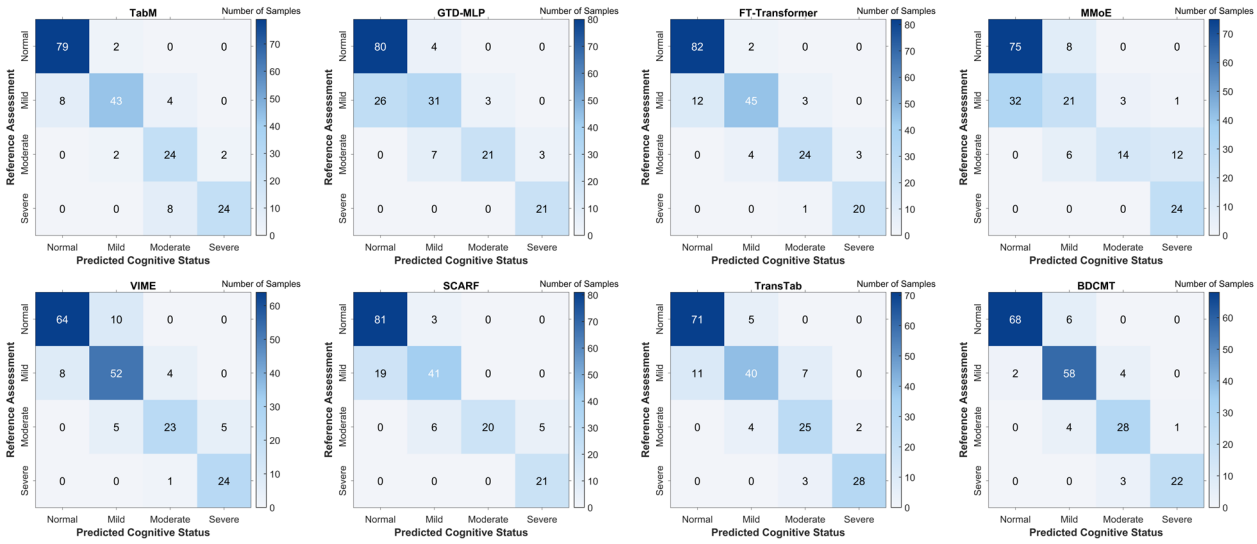


Fig. 7 Confusion matrix of different models in cognitive function grading tasks.

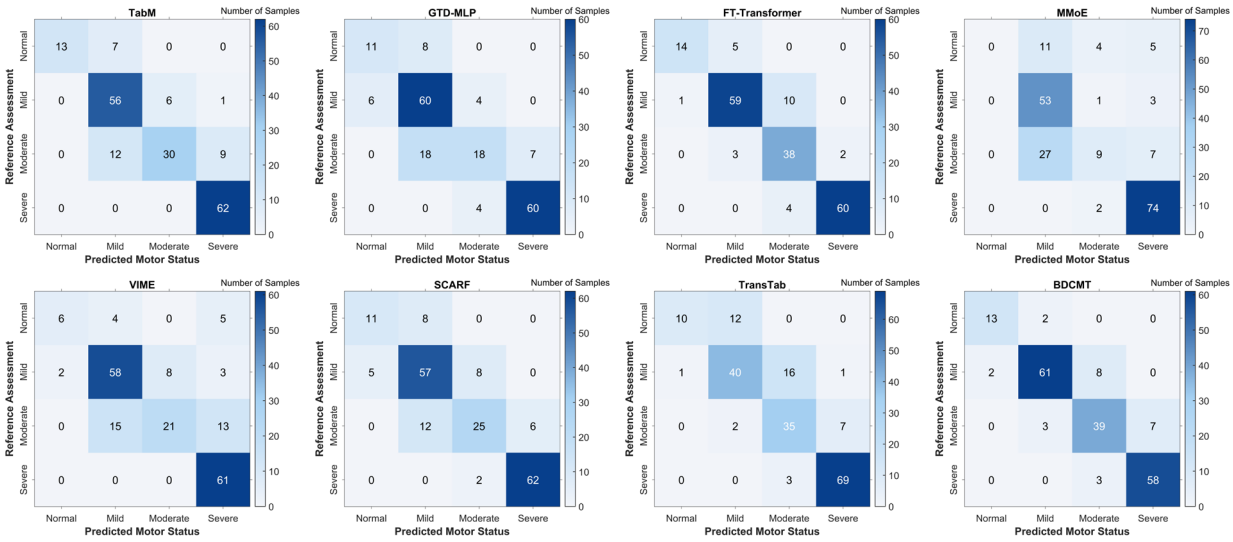


Fig. 8 Confusion matrix of different models in motor function grading task.

Fig. 9 compares the execution times of different models. Lightweight MLP-based and self-supervised baselines generally required less time, whereas Transformer-based models and transfer-oriented tabular models introduced higher computational overhead. BDCMT required additional computation because it includes masked feature reconstruction pre-training, dual-stream Transformer encoding, reliability-gated bidirectional cross-attention, and distributional ordinal loss. Nevertheless, its execution time remained within an acceptable range and was lower than the most computationally expensive Transformer-based baseline. These results suggest that BDCMT achieves a reasonable balance between improved grading performance and computational cost.

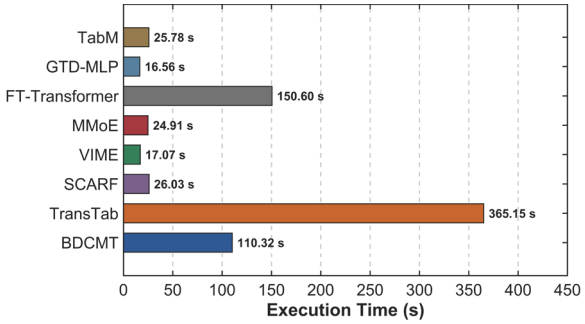


Fig. 9 Comparison of execution time for different models.

4.2.2 Statistical Significance Analysis

To further evaluate the performance differences between models, the Wilcoxon signed-rank test was used to compare BDCMT with each baseline model based on the paired results from the same ten repeated runs, with $p < 0.05$ considered statistically significant. The Friedman test was then used to evaluate the overall performance differences among the eight models.

As shown in **Table 3**, statistically significant differences were observed between BDCMT and all comparison models across the five-evaluation metrics in both cognitive and motor impairment grading tasks. All p-values were below 0.05. For cognitive grading, the p-values ranged from $1.95 \times 10{-}3$ to $1.95 \times 10{-}2$, whereas for motor grading, they ranged from $1.95 \times 10{-}3$ to $7.81 \times 10{-}3$. Together with the direction of the

mean performance differences reported in **Table 2**, these paired comparisons support the performance advantages of BDCMT across repeated runs.

Table 3 Wilcoxon signed-rank test results between BDCMT and comparison models.

Category	Metric	BDCMT VS						
		TabM	GTD-MLP	FT-Transformer	MMoE	VIME	SCARF	TransTab
Cognitive Impairment	ACC	1.95E-02	1.95E-03	7.81E-03	1.95E-03	1.95E-03	3.91E-03	1.37E-02
	AUC_{OVR}	3.91E-03	1.95E-03	1.95E-03	1.95E-03	1.95E-03	1.95E-03	1.95E-03
	F1(w)	9.77E-03	1.95E-03	9.77E-03	1.95E-03	1.95E-03	1.95E-03	1.37E-02
	MCC	1.37E-02	1.95E-03	5.86E-03	1.95E-03	1.95E-03	1.95E-03	1.37E-02
	Grade MAE	1.56E-02	1.95E-03	7.81E-03	1.95E-03	1.95E-03	3.91E-03	1.37E-02
Motor Impairment	ACC	1.95E-03	1.95E-03	7.81E-03	1.95E-03	1.95E-03	1.95E-03	1.95E-03
	AUC_{OVR}	1.95E-03	1.95E-03	1.95E-03	1.95E-03	1.95E-03	1.95E-03	1.95E-03
	F1(w)	1.95E-03	1.95E-03	1.95E-03	1.95E-03	1.95E-03	1.95E-03	1.95E-03
	MCC	1.95E-03	1.95E-03	1.95E-03	1.95E-03	1.95E-03	1.95E-03	1.95E-03
	Grade MAE	1.95E-03	1.95E-03	7.81E-03	1.95E-03	1.95E-03	1.95E-03	1.95E-03

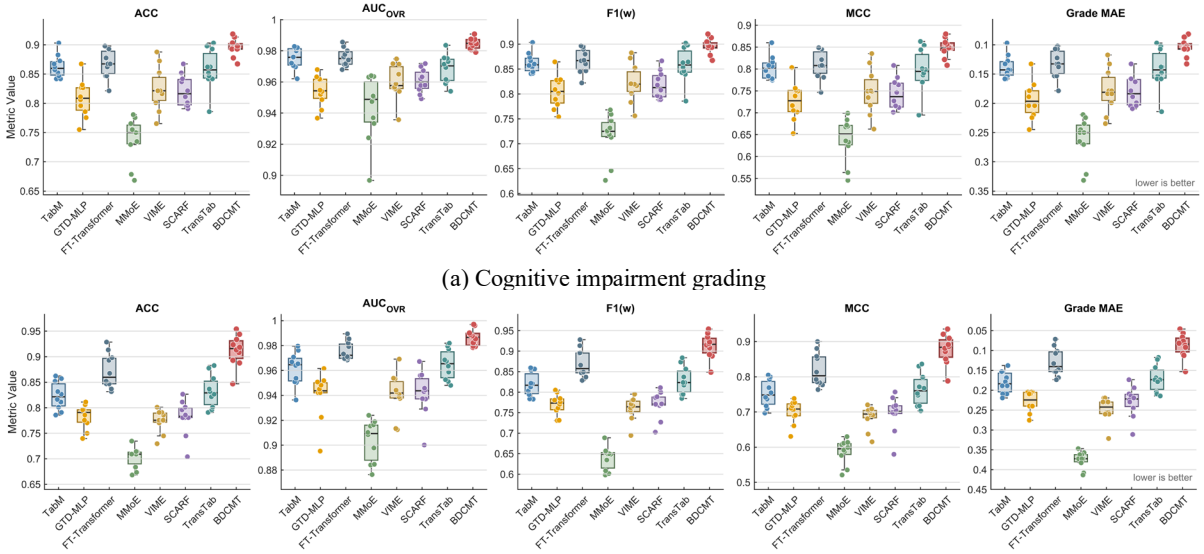
For each task, the Friedman test was performed using the results from the ten repeated runs, and **Table 4** reports the mean rank of each model across these runs. BDCMT achieved the best average ranking in both cognitive and motor scoring tasks, ranking first in both. FT-Transformer ranked second in both tasks, while TabM ranked third. TransTab ranked relatively better in the motor scoring task than in the cognitive scoring task. Overall, the results of statistical significance analysis indicate that BDCMT has the strongest overall performance among the compared models, which is statistically significant.

Table 4 Friedman test results for different models.

Algorithm	Mean Rank		Rank	
	Cognitive Impairment	Motor Impairment	Cognitive Impairment	Motor Impairment
TabM	2.66	3.92	3	3
GTD-MLP	5.76	5.56	7	6
FT-Transformer	2.36	2.00	2	2
MMoE	7.86	8.00	8	8
VIME	5.43	5.94	5	7
SCARF	5.38	5.42	4	5
TransTab	5.55	4.16	6	4
BDCMT	1.00	1.00	1	1

Fig. 10 utilizes boxplots [43] to present the distribution of model performance across repeated runs for the cognitive and motor impairment grading tasks. BDCMT consistently achieved the best overall results, with higher ACC, AUC_{OVR}, F1(w), and MCC, and lower Grade MAE than the competing methods. The narrower distributions of BDCMT also suggest good run-to-run stability. Although FT-Transformer, TabM,

and TransTab provided competitive baselines on several metrics, BDCMT maintained superior performance across most metrics and both tasks. These results further support the effectiveness and stability of reliability-gated bidirectional cross-domain interaction and distributional ordinal loss for joint cognitive and motor impairment grading.



(a) Cognitive impairment grading

(b) Motor impairment grading

Fig. 10 Boxplots of model performance across repeated runs for cognitive and motor impairment grading.

4.2.3 Ablation Studies

The ablation experiments in **Fig. 11** show that the complete BDCMT achieved the best overall performance in both cognitive and motor impairment grading tasks. Removing the reliability-gated bidirectional cross-attention module reduced performance across multiple classification metrics and increased grade prediction error, indicating that controlled cognitive-motor information exchange is important for joint assessment. Removing masked feature reconstruction or feature-specific domain-aware tokenization also reduced model performance, suggesting that these components improve structured feature representation and domain-specific information modeling. Removing the distributional ordinal loss led to an evident decrease in ordinal grading performance, especially in terms of grade-distance-related error. This indicates that distributional ordinal constraints help reduce clinically undesirable distant-grade misclassification. Overall, all components contribute positively to the performance of BDCMT, with the reliability-gated bidirectional cross-attention and distributional ordinal loss being particularly significant in their contributions to the final performance.

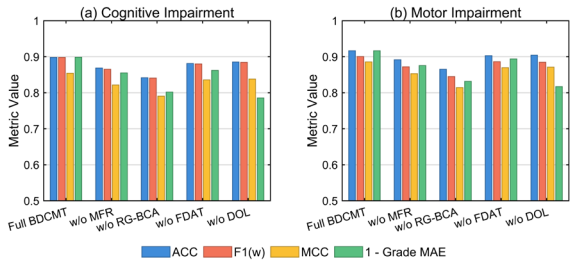


Fig. 11 Ablation study of the proposed BDCMT on (a) cognitive impairment and (b) motor impairment grading tasks. Full BDCMT denotes the complete model, whereas w/o MFR, w/o RG-BCA, w/o FDAT, and w/o DOL denote the variants without masked feature reconstruction, reliability-gated bidirectional cross-attention, feature-specific domain-aware tokenization, and distributional ordinal loss, respectively.

4.3 Evaluation of LLM-Based Decision-Support Configurations

After evaluating the cognitive-motor assessment stage, we examined whether the resulting structured functional profile could support personalized rehabilitation decision support. This part of the experiment focused on the downstream assessment-to-decision workflow. Three LLM configurations were compared using the same 40 stratified cases: Base-LLM, RAG-LLM, and QC-RAG-LLM. To ensure a controlled comparison, all three configurations used the same deployed language model, decoding settings, scale-interpretation rules, safety constraints, terminology requirements, output structure, and stop-training warnings. These shared instructions constrained the interpretation of MMSE, MoCA-B, SPPB, grip strength, TUG, and fall history; required appropriate supervision and fall-prevention measures; standardized the five-part report structure; and required training to be stopped and medical evaluation to be sought when predefined warning symptoms occurred.

The three configurations differed in the assessment and evidence information supplied to the common generation framework. Base-LLM received the original demographic, cognitive, and motor assessment information. RAG-LLM received the same information together with retrieved clinical evidence. QC-RAG-LLM additionally received the BDCMT-derived cognitive and motor grades, class-probability distributions, deterministic scale summaries, and attention-derived relevance cues in the form of a structured functional profile. Thus, the comparison evaluated whether retrieval and structured quantitative conditioning provided additional value under the same basic scale, safety, terminology, and formatting controls.

In addition to these shared generation controls, QC-RAG-LLM adopted a prompt-constrained generation strategy to further preserve assessment consistency. The report title, overall assessment conclusion, and the first assessment section were generated programmatically from the structured functional profile and provided to the LLM as fixed context. The LLM was only responsible for generating the intervention-related sections, including multidimensional cognitive rehabilitation training, personalized motor intervention, home safety protection, and periodic reassessment with dynamic adjustment. In addition, the prompt constrained scale interpretation, terminology usage, exercise intensity, fall-history description, and stop-training warnings.

The program-fixed assessment section was only applied to QC-RAG-LLM. Base-LLM and RAG-LLM

followed the same five-part report structure but generated the assessment-related content directly from their respective input information without this fixed assessment context. Therefore, the comparison reflects the performance differences among the three complete configurations rather than an isolated causal evaluation of structured quantitative conditioning. The detailed input configurations, quantitative conditioning components, and prompt-level constraints of the three LLM-based settings are summarized in **Table 5**.

Table 5 Input configurations and prompt-level constraints of the three LLM-based settings.

Component	Applied to	Description
Common generation controls	All three settings	Same LLM, decoding settings, scale-interpretation rules, safety constraints, terminology requirements, report structure, and stop-training warnings
Original assessment information	All three settings	Demographic information and original cognitive-motor measurements
Retrieved clinical evidence	RAG-LLM and QC-RAG-LLM	Cognitive and motor rehabilitation knowledge retrieved from domain-specific knowledge bases
Structured functional profile	QC-RAG-LLM	BDCMT-derived grades, class-probability distributions, deterministic scale summaries, and attention-derived relevance cues
Program-fixed assessment section	QC-RAG-LLM	Report title, overall assessment conclusion, and first assessment section generated from the structured functional profile and provided as fixed context
Programmatic output validation	QC-RAG-LLM	Required headings, prohibited terminology, motor-safety conditions, and stop-training statements were checked; one corrective regeneration was allowed when validation failed

The generated personalized recommendations were evaluated by two clinicians using the S.C.O.R.E. framework [44], which includes safety, consensus, objectivity, reproducibility, and explainability [45]. Each dimension was rated on a five-point Likert scale, with higher scores indicating better clinical acceptability. The overall S.C.O.R.E. value was calculated as the average of the five dimensions for each case.

As shown in **Table 6**, QC-RAG-LLM obtained the highest mean scores across all five dimensions and the overall S.C.O.R.E. value. RAG-LLM improved consensus and explainability compared with Base-LLM, suggesting that retrieved clinical literature helped align the generated recommendations with clinical knowledge and provided more explicit supporting rationale. However, RAG-LLM showed limited improvement in safety and reproducibility. In contrast, QC-RAG-LLM achieved higher scores in safety, consensus, reproducibility, and explainability, indicating that quantitative conditioning based on BDCMT outputs and prompt-level constraints provided additional clinical value beyond retrieval alone.

Table 6 Clinician-rated S.C.O.R.E. results of LLM variants (mean ± SD).

	Safety	Consensus	Objectivity	Reproducibility	Explainability	Overall SCORE
Base-LLM	2.50 ± 0.64	2.33 ± 0.66	4.03 ± 0.42	2.88 ± 0.88	1.10 ± 0.30	2.56 ± 0.34
RAG-LLM	2.40 ± 0.74	2.75 ± 0.63	3.83 ± 0.59	3.00 ± 0.88	3.95 ± 0.50	3.18 ± 0.42
QC-RAG-LLM	4.88 ± 0.40	4.47 ± 0.51	4.83 ± 0.38	4.75 ± 0.44	4.47 ± 0.51	4.68 ± 0.21

Fig. 12 displays the five-dimensional S.C.O.R.E. radar chart for these LLM variants. QC-RAG-LLM showed a more balanced profile and higher mean scores across the evaluated dimensions. **Fig. 13** further shows the case-level distribution and paired changes of the overall S.C.O.R.E. value, utilizing a violin plot [46] and a paired line plot to illustrate the density and individual trajectories, respectively. The paired trajectories indicate a general increase from Base-LLM to RAG-LLM and a further increase from RAG-LLM to QC-RAG-LLM in most cases. This trend supports the incremental contribution of retrieval-grounded generation and quantitative assessment conditioning.

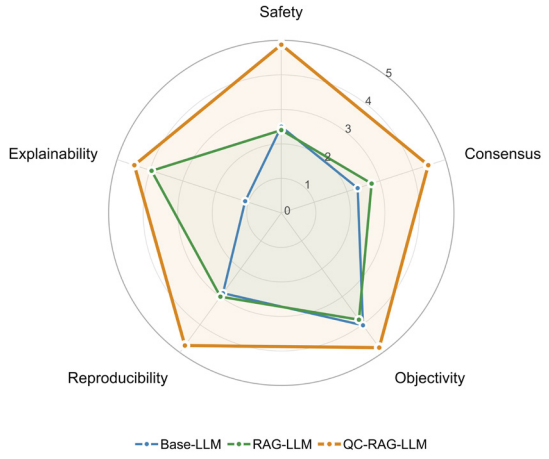


Fig. 12 Five-dimensional S.C.O.R.E. radar chart of LLM variants.

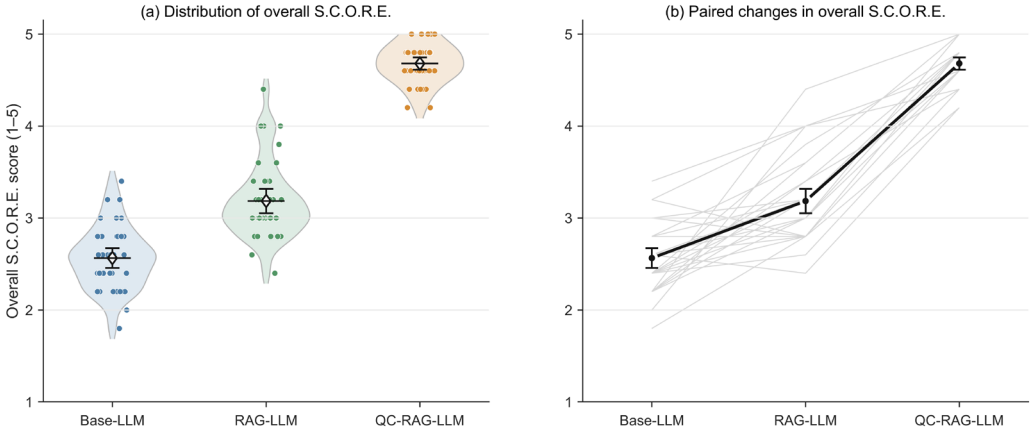


Fig. 13 Overall S.C.O.R.E. distribution and paired changes.

Paired nonparametric tests were used to compare the clinician-assessed S.C.O.R.E. scores of the three LLM models. For each score, the Friedman test was first applied to assess the overall differences between Base-LLM, RAG-LLM, and QC-RAG-LLM in the same 40 cases. Effect sizes are expressed as Kendall's W [47]. As shown in **Table 7**, significant differences were observed among the three variants for the overall S.C.O.R.E. value and all five dimensions. The corresponding Kendall's W values indicated moderate-to-large effect sizes, and QC-RAG-LLM ranked first across all evaluated outcomes, indicating excellent performance.

Post-hoc pairwise comparisons were then performed to further examine whether QC-RAG-LLM differed from the two reference variants. Two-sided Wilcoxon signed-rank tests were used, and Holm correction was applied to the two pre-specified comparisons for each outcome. As shown in **Table 8**, QC-RAG-LLM significantly outperformed both Base-LLM and RAG-LLM in the overall S.C.O.R.E. score and all five dimensions, with all Holm-adjusted p values below 0.001. The rank-biserial correlations were large and directionally consistent, further supporting the advantage of QC-RAG-LLM in clinician-rated personalized rehabilitation recommendations quality.

Table 7 Friedman tests for clinician-rated S.C.O.R.E. outcomes.

Outcome	Indicator	Base-LLM	RAG-LLM	QC-RAG-LLM	Friedman χ^2	p-value	Kendall's W
Overall S.C.O.R.E.	Rank	3	2	1	73.72	<0.001	0.92
Safety	Rank	2	3	1	66.96	<0.001	0.84
Consensus	Rank	3	2	1	65.19	<0.001	0.82
Objectivity	Rank	2	3	1	53.58	<0.001	0.67
Reproducibility	Rank	3	2	1	53.80	<0.001	0.67
Explainability	Rank	3	2	1	72.18	<0.001	0.90

Table 8 Wilcoxon signed-rank tests for clinician-rated S.C.O.R.E. outcomes.

Outcome	QC-RAG-LLM vs Base-LLM Holm-adjusted *P*	Rank-biserial *r*	QC-RAG-LLM vs RAG-LLM Holm-adjusted *P*	Rank-biserial *r*
Overall S.C.O.R.E.	<0.001	1.00	<0.001	1.00
Safety	<0.001	1.00	<0.001	1.00
Consensus	<0.001	1.00	<0.001	1.00
Objectivity	<0.001	1.00	<0.001	1.00
Reproducibility	<0.001	1.00	<0.001	0.98
Explainability	<0.001	1.00	<0.001	0.84

Fig. 14 compares the execution times of the three LLM variants across three independent runs. RAG-LLM had the longest execution time and the greatest inter-run variation, which is related to the retrieval and generation processes. In contrast, the execution time of QC-RAG-LLM is close to that of Base-LLM without RAG. This suggests that the additional computational overhead introduced by the proposed quantitative conditions and cue-level constraints is limited. Combined with the S.C.O.R.E. evaluation results, these results indicate that QC-RAG-LLM improves the clinical quality of rehabilitation generation plans without significantly affecting execution efficiency.

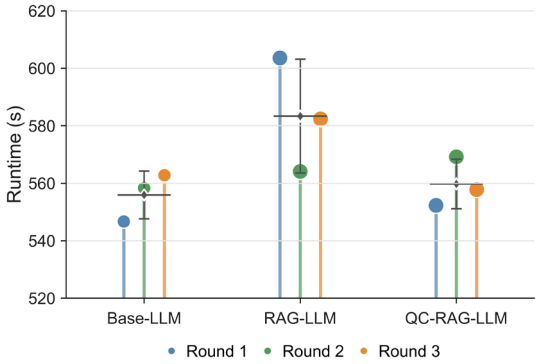


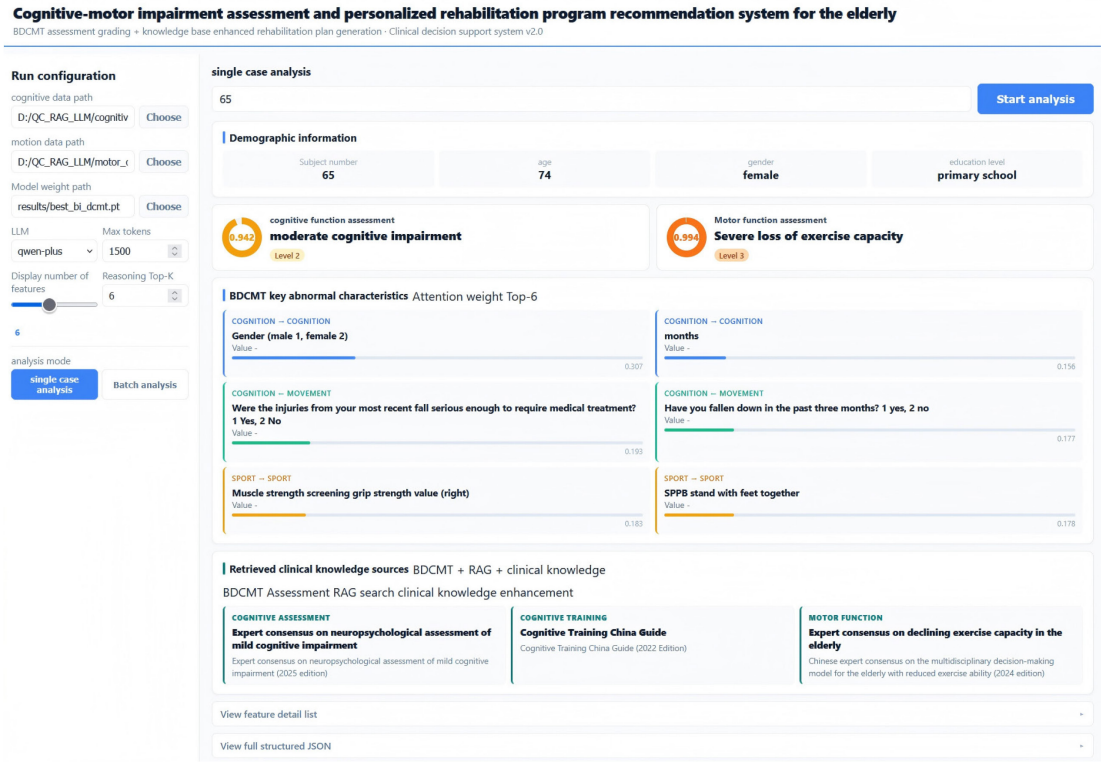
Fig. 14 Runtime comparison of LLM variants. Colored lollipop markers show individual runs, while black diamonds and error bars indicate mean ± SD.

4.4 System Demonstration

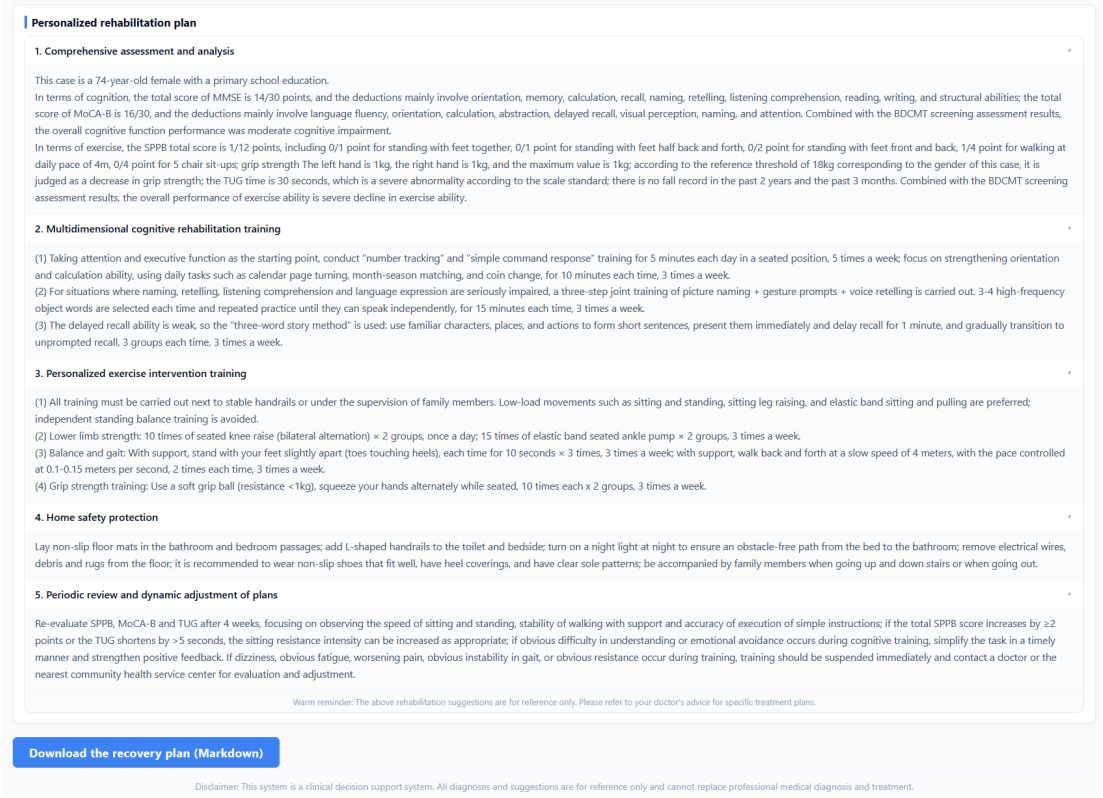
Fig. 15 presents the web-based decision-support system implementing the proposed assessment-to-decision workflow. The interface maintains the structured functional profile across cognitive-motor grading, evidence retrieval, and personalized rehabilitation recommendation generation, allowing quantitative assessment results and retrieved clinical evidence to be reviewed before the final plan is confirmed. As shown in **Fig. 15 (a)**, the main interface displays demographic information, cognitive and motor assessment results, BDCMT-derived grading outputs, attention-derived relevance cues, and retrieved clinical evidence. These components provide a structured quantitative basis and an evidence-grounded context for generating personalized recommendations.

Fig. 15 (b) shows a representative rehabilitation recommendation generated by QC-RAG-LLM. The plan begins with a program-fixed assessment summary and then provides personalized cognitive training, motor intervention, home safety precautions, and follow-up adjustment recommendations. In this case, cognitive training is linked to the MoCA-B and MMSE domains with score deductions, while motor intervention is adjusted according to the BDCMT motor grade, SPPB score, grip strength, and TUG result. The generated recommendations also specify practical safety boundaries, including seated or supported exercise, caregiver accompaniment, home anti-slip measures, and symptom-triggered suspension of training.

This demonstration shows how the structured functional profile is maintained across cognitive-motor assessment, evidence retrieval, and personalized rehabilitation recommendation generation within a single decision-support workflow. Instead of producing an isolated risk label or a generic text response, the system converts structured assessment results into a structured, evidence-informed rehabilitation recommendation for clinician review. Such an assessment-to-decision workflow may help support community-based screening, clinician review, rehabilitation guidance, and follow-up management for older adults with cognitive-motor impairment.



(a) Main interface of the proposed system



(b) Representative output of the personalized rehabilitation recommendation

Fig. 15 Web-based clinical decision-support system for cognitive-motor impairment assessment and personalized rehabilitation decision support.

5. Discussion

This study investigates cognitive-motor assessment and rehabilitation decision support as a connected information processing process rather than two independent tasks. The proposed BDCMT effectively integrates heterogeneous cognitive and motor measurements by preserving domain-specific representations and enabling controlled cross-domain information exchange. The improved performance in both cognitive and motor impairment grading, particularly the larger improvement observed in motor assessment, suggests that complementary information between functional domains can contribute to more comprehensive characterization of impairment status. These findings support the use of reliability-gated cross-domain representation learning for heterogeneous clinical information fusion. Furthermore, the distributional ordinal loss incorporates the ordered structure of impairment severity, allowing the model to better distinguish clinically adjacent and distant impairment levels.

A key aspect of the proposed framework is that assessment results are not treated as the final output of the system. Conventional prediction models often compress multidimensional clinical information into a single label or probability score, which may limit their usefulness for subsequent decision support. In this study, the structured functional profile preserves predicted grades, probability distributions, original measurements, deterministic scale summaries, demographic information, and attention-derived relevance cues. This representation provides an intermediate quantitative interface between assessment and downstream evidence retrieval, allowing clinically meaningful information to be retained throughout the decision-support process. Rather than replacing assessment results with free-text descriptions, the proposed approach maintains a structured connection between quantitative evaluation and personalized rehabilitation recommendations.

The clinician evaluation further demonstrates the potential value of integrating quantitative assessment information with knowledge-grounded generation. Although retrieval-augmented generation provides access to external rehabilitation knowledge, retrieved evidence must still be aligned with individual functional characteristics. QC-RAG-LLM uses the structured functional profile to condition evidence retrieval and generation while incorporating predefined safety constraints. The higher clinician ratings across the S.C.O.R.E. dimensions indicate the potential benefit of combining structured assessment information, retrieved evidence, and language generation within a unified decision-support framework. These findings should be interpreted as an evaluation of the complete workflow rather than an isolated assessment of the language model component. The LLM serves as a tool for organizing quantitative information and supporting evidence into interpretable recommendations for clinician review, rather than independently determining functional status or rehabilitation decisions.

The proposed framework may provide a practical approach for community-based screening and follow-up scenarios, where cognitive and motor measurements are available but require integration and interpretation. By preserving intermediate representations and retrieved evidence, the framework improves

the traceability of the decision-support process and allows clinicians to examine the basis of generated recommendations before making final decisions. Therefore, the system should be considered a clinical decision-support tool rather than an autonomous diagnostic or treatment system.

However, this study has some limitations. While the dataset comes from older adults in multiple communities within the same cohort, external validation in a broader population is still needed. The assessment of the rehabilitation recommendation was based on clinician ratings, not prospective intervention outcomes, so the clinical validity of the generated recommendations remains to be determined. Furthermore, attention-derived relevance cues should be interpreted as model-level indicators of feature relevance, rather than causal explanations.

6. Conclusion

This study developed a reliability-gated cross-domain information fusion framework for cognitive-motor assessment and personalized rehabilitation decision support in older adults. The proposed BDCMT preserves cognitive and motor domain characteristics through separate representation streams, regulates complementary information exchange using reliability-gated bidirectional cross-attention, and incorporates the ordered structure of impairment severity through a distributional ordinal loss. Experiments conducted on 977 community-dwelling older adults achieved mean ACC values of 0.8867 and 0.9036 for cognitive and motor impairment grading, respectively.

This framework does not use predicted grades as the final output. Instead, it integrates assessment results, class-probability distributions, original measurements, deterministic scale summaries, demographic information, and attention-derived relevance cues into a structured functional profile. This profile provides a quantitative interface between assessment and rehabilitation decision support, supporting domain-specific evidence retrieval and the generation of plans with safety constraints. In clinician assessments, the complete QC-RAG-LLM configuration achieved the highest overall S.C.O.R.E. score. These results support the feasibility of retaining quantitative assessment information throughout the assessment-to-decision workflow.

Overall, the proposed framework provides a traceable connection between quantitative cognitive-motor assessment, evidence retrieval, and personalized rehabilitation decision support. It provides a potential approach for community-based assessment, clinician review, and follow-up management by preserving clinically relevant information throughout the decision-support process. Future work will focus on external validation [48], prospective clinical evaluation [49], integration with closed-loop rehabilitation training robots [50], and broader testing across different knowledge bases and LLMs [51,52].

References

[1] G. Livingston, J. Huntley, K.Y. Liu, S.G. Costafreda, G. Selbæk, S. Alladi, D. Ames, S. Banerjee, A. Burns, C. Brayne, N.C. Fox, C.P. Ferri, L.N. Gitlin, R. Howard, H.C. Kales, M. Kivimäki, E.B. Larson, N. Nakasujja, K. Rockwood, Q. Samus, K. Shirai, A. Singh-Manoux, L.S. Schneider, S. Walsh, Y. Yao, A. Sommerlad, N. Mukadam, Dementia prevention, intervention, and care: 2024 report of the Lancet standing Commission, The Lancet 404 (2024) 572–628. https://doi.org/10.1016/S0140-6736(24)01296-0.

[2] C. Maiocchi, M. Milanesi, N. Canessa, A. Nardone, C. Gianelli, Effects of cognitive-motor interference during walking: A meta-analysis on healthy individuals, Neurosci. Biobehav. Rev. 180 (2026) 106490. https://doi.org/10.1016/j.neubiorev.2025.106490.

[3] Y. Hao, Y. Zhao, H. Luo, L. Xie, H. Hu, C. Sun, Comparative effectiveness of different dual task mode interventions on cognitive function in older adults with mild cognitive impairment or dementia: a systematic review and network meta-analysis, Aging Clin. Exp. Res. 37 (2025) 139. https://doi.org/10.1007/s40520-025-03016-5.

[4] M. Montero-Odasso, N. van der Velde, F.C. Martin, M. Petrovic, M.P. Tan, J. Ryg, S. Aguilar-Navarro, N.B. Alexander, C. Becker, H. Blain, R. Bourke, I.D. Cameron, R. Camicioli, L. Clemson, J. Close, K. Delbaere, L. Duan, G. Duque, S.M. Dyer, T. ... Masud, World guidelines for falls prevention and management for older adults: A global initiative, Age Ageing 51 (2022) afac205. https://doi.org/10.1093/ageing/afac205.

[5] E. Ayers, C. Wang, J. Verghese, Validation of a “subjective motoric cognitive risk syndrome” screening tool for motoric cognitive risk syndrome—A prospective cohort study, Eur. J. Neurol. 29 (2022) 2925–2933. https://doi.org/10.1111/ene.15476.

[6] S. Wiegrebe, P. Kopper, R. Sonabend, B. Bischl, A. Bender, Deep learning for survival analysis: a review, Artif. Intell. Rev. 57 (2024) 65. https://doi.org/10.1007/s10462-023-10681-3.

[7] F. Krones, U. Marikkar, G. Parsons, A. Szmul, A. Mahdi, Review of multimodal machine learning approaches in healthcare, Inf. Fusion 114 (2025) 102690. https://doi.org/10.1016/j.inffus.2024.102690.

[8] L. Wang, H. Wang, Y. Su, F. Lure, J. Li, A Novel Hybrid Ordinal Learning Model With Health Care Application, IEEE Trans. Autom. Sci. Eng. 22 (2025) 339–352. https://doi.org/10.1109/TASE.2024.3350894.

[9] C. Wang, Y. Lei, T. Chen, J. Zhang, Y. Li, H. Shan, HOPE: Hybrid-Granularity Ordinal Prototype Learning for Progression Prediction of Mild Cognitive Impairment, IEEE J. Biomed. Health Inform. 28 (2024) 6429–6440. https://doi.org/10.1109/JBHI.2024.3357453.

[10] M.M. Rahman, M.E. Masry, S.C. Gnyawali, Y. Xue, G. Gordillo, J.P. Wachs, AI-Driven Integrated System for Burn Depth Prediction With Electronic Medical Records: Algorithm Development and

Validation, JMIR Med. Inform. 13 (2025) e68366–e68366. https://doi.org/10.2196/68366.

[11] P.D. Manage, Y. Wu, J. Zhang, A.U. Gunathilaka, Y. Li, Overcoming BERT's limitations in uncertainty: A novel two-stage solution for multi-class medical text classification, Knowl.-Based Syst. 336 (2026) 115315. https://doi.org/10.1016/j.knosys.2026.115315.

[12] Z. Zhao, W. Fan, J. Li, Y. Liu, X. Mei, Y. Wang, Z. Wen, F. Wang, X. Zhao, J. Tang, Q. Li, Recommender Systems in the Era of Large Language Models (LLMs), IEEE Trans. Knowl. Data Eng. 36 (2024) 6889–6907. https://doi.org/10.1109/TKDE.2024.3392335.

[13] D. Van Veen, C. Van Uden, L. Blankemeier, J.-B. Delbrouck, A. Aali, C. Bluethgen, A. Pareek, M. Polacin, E.P. Reis, A. Seehofnerová, N. Rohatgi, P. Hosamani, W. Collins, N. Ahuja, C.P. Langlotz, J. Hom, S. Gatidis, J. Pauly, A.S. Chaudhari, Adapted large language models can outperform medical experts in clinical text summarization, Nat. Med. 30 (2024) 1134–1142. https://doi.org/10.1038/s41591-024-02855-5.

[14] A. Retevoi, G. Devittori, T. Kowatsch, O. Lambercy, Building Conversational Agents for Stroke Rehabilitation: An Evaluation of Large Language Models and Retrieval Augmented Generation, in: Proc. 24th ACM Int. Conf. Intell. Virtual Agents, Association for Computing Machinery, New York, NY, USA, 2024: pp. 1–4. https://doi.org/10.1145/3652988.3673924.

[15] S. Kresevic, M. Giuffrè, M. Ajcevic, A. Accardo, L.S. Crocè, D.L. Shung, Optimization of hepatological clinical guidelines interpretation by large language models: a retrieval augmented generation-based framework, Npj Digit. Med. 7 (2024) 102. https://doi.org/10.1038/s41746-024-01091-y.

[16] F. Di Martino, F. Delmastro, Explainable AI for clinical and remote health applications: a survey on tabular and time series data, Artif. Intell. Rev. 56 (2022) 5261–5315. https://doi.org/10.1007/s10462-022-10304-3.

[17] L. Grinsztajn, E. Oyallon, G. Varoquaux, Why do tree-based models still outperform deep learning on typical tabular data? Advances in Neural Information Processing Systems, 35, 507-520., (2022).

[18] S.Ö. Arik, T. Pfister, TabNet: Attentive Interpretable Tabular Learning, Proc. AAAI Conf. Artif. Intell. 35 (2021) 6679–6687. https://doi.org/10.1609/aaai.v35i8.16826.

[19] Y. Gorishniy, A. Kotelnikov, A. Babenko, TabM: Advancing tabular deep learning with parameter-efficient ensembling, Int. Conf. Learn. Represent. 2025 (2025) 77899–77935.

[20] J. Yoon, Y. Zhang, J. Jordon, M. van der Schaar, VIME: Extending the Success of Self- and Semi-supervised Learning to Tabular Domain, in: Adv. Neural Inf. Process. Syst., Curran Associates, Inc., 2020: pp. 11033–11043. https://proceedings.neurips.cc/paper_files/paper/2020/hash/7d97667a3e056acab9aaf653807b4a03-Abstract.html (accessed November 30, 2025).

[21] D. Bahri, H. Jiang, Y. Tay, D. Metzler, SCARF: Self-supervised contrastive learning using random

feature corruption, in: International Conference on Learning Representations (ICLR), 2022. https://openreview.net/forum?id=CuV_qYkmKb3.

[22] Z. Wang, J. Sun, Transtab: Learning transferable tabular transformers across tables, Advances in Neural Information Processing Systems 35 (2022) 2902–2915.

[23] J. Ma, Z. Zhao, X. Yi, J. Chen, L. Hong, E.H. Chi, Modeling Task Relationships in Multi-task Learning with Multi-gate Mixture-of-Experts, in: Proc. 24th ACM SIGKDD Int. Conf. Knowl. Discov. Data Min., Association for Computing Machinery, New York, NY, USA, 2018: pp. 1930–1939. https://doi.org/10.1145/3219819.3220007.

[24] D. Jung, J. Kim, M. Kim, C.W. Won, K.-R. Mun, Classifying the Risk of Cognitive Impairment Using Sequential Gait Characteristics and Long Short-Term Memory Networks, IEEE J. Biomed. Health Inform. 25 (2021) 4029–4040. https://doi.org/10.1109/JBHI.2021.3073372.

[25] B. Ghoraani, L.N. Boettcher, M.D. Hssayeni, A. Rosenfeld, M.I. Tolea, J.E. Galvin, Detection of mild cognitive impairment and Alzheimer's disease using dual-task gait assessments and machine learning, Biomed. Signal Process. Control 64 (2021) 102249. https://doi.org/10.1016/j.bspc.2020.102249.

[26] X. Wei, C. Huang, X. Ding, Z. Zhou, Y. Zhang, X. Feng, S. Zheng, T. Li, J. Lü, Effect of virtual reality training on dual-task performance in older adults: a systematic review and meta-analysis, J. NeuroEngineering Rehabil. 22 (2025) 141. https://doi.org/10.1186/s12984-025-01675-z.

[27] Y. Gao, N. Liu, Effectiveness of cognitive-motor dual task training in preventing falls in community older adults: A meta-analysis and systematic review, Geriatr. Nur. 64 (2025) 103366. https://doi.org/10.1016/j.gerinurse.2025.05.005.

[28] R. Kleinlein, D.W. Bates, C. Guan, K.J. Gray, V.P. Kovacheva, SPELL: A scalable NLP method using regular expressions and large language models for clinical information extraction, Comput. Methods Programs Biomed. 285 (2026) 109517. https://doi.org/10.1016/j.cmpb.2026.109517.

[29] Y. Wu, T. Chen, I. Hochberg, Z. Sun, R. Edry, Z. Huang, M. Peleg, Beyond Accuracy: Safety-Centered guidelines for the evaluation of LLM-based therapy recommendation systems for chronic multimorbidity patients, J. Biomed. Inform. 180 (2026) 105072. https://doi.org/10.1016/j.jbi.2026.105072.

[30] M. Jeong, J. Sohn, M. Sung, J. Kang, Improving medical reasoning through retrieval and self-reflection with retrieval-augmented large language models, Bioinformatics 40 (2024) i119–i129. https://doi.org/10.1093/bioinformatics/btae238.

[31] Y.S. Low, M.L. Jackson, R.J. Hyde, R.E. Brown, N.M. Sanghavi, J.D. Baldwin, C.W. Pike, J. Muralidharan, G. Hui, N. Alexander, H. Hassan, R.V. Nene, M. Pike, C.J. Pokrzywa, S. Vedak, A.P. Yan, D.H. Yao, A.R. Zipursky, C. Dinh, S. ... Gombar, Answering real-world clinical questions using large language model, retrieval-augmented generation, and agentic systems, Digit. Health 11 (2025)

20552076251348850. https://doi.org/10.1177/20552076251348850.

[32] B.B. Ozmen, P. Mathur, Evidence-based artificial intelligence: Implementing retrieval-augmented generation models to enhance clinical decision support in plastic surgery, J. Plast. Reconstr. Aesthet. Surg. 104 (2025) 414–416. https://doi.org/10.1016/j.bjps.2025.03.053.

[33] H. Zou, Y. Wang, Large language model augmented framework with domain-specific knowledge integration for medical named entity recognition, Neural Netw. 203 (2026) 109178. https://doi.org/10.1016/j.neunet.2026.109178.

[34] B. Vasey, S. Ursprung, B. Beddoe, E.H. Taylor, N. Marlow, N. Bilbro, P. Watkinson, P. McCulloch, Association of Clinician Diagnostic Performance With Machine Learning–Based Decision Support Systems: A Systematic Review, JAMA Netw. Open 4 (2021) e211276. https://doi.org/10.1001/jamanetworkopen.2021.1276.

[35] M. Ghassemi, L. Oakden-Rayner, A.L. Beam, The false hope of current approaches to explainable artificial intelligence in health care, Lancet Digit. Health 3 (2021) e745–e750. https://doi.org/10.1016/S2589-7500(21)00208-9.

[36] M. Etemadi, S. Bazzaz Abkenar, A. Ahmadzadeh, M. Haghi Kashani, P. Asghari, M. Akbari, E. Mahdipour, A systematic review of healthcare recommender systems: Open issues, challenges, and techniques, Expert Syst. Appl. 213 (2023) 118823. https://doi.org/10.1016/j.eswa.2022.118823.

[37] J. Clusmann, F.R. Kolbinger, H.S. Muti, Z.I. Carrero, J.-N. Eckardt, N.G. Laleh, C.M.L. Löffler, S.-C. Schwarzkopf, M. Unger, G.P. Veldhuizen, S.J. Wagner, J.N. Kather, The future landscape of large language models in medicine, Commun. Med. 3 (2023) 141. https://doi.org/10.1038/s43856-023-00370-1.

[38] K. Han, Y. Wang, H. Chen, X. Chen, J. Guo, Z. Liu, Y. Tang, A. Xiao, C. Xu, Y. Xu, Z. Yang, Y. Zhang, D. Tao, A Survey on Vision Transformer, IEEE Trans. Pattern Anal. Mach. Intell. 45 (2023) 87–110. https://doi.org/10.1109/TPAMI.2022.3152247.

[39] S. Cheng, J. Qin, C. Hou, Y. Wu, X. Du, H. Liu, S. Lei, R. Li, X. Yue, Y. Guo, Linking Cognitive Screening Tests in Community-Dwelling Older Adults: Crosswalk between the Montreal Cognitive Assessment-Basic and the Mini-Mental State Examination, J. Am. Med. Dir. Assoc. 26 (2025) 105550. https://doi.org/10.1016/j.jamda.2025.105550.

[40] M.S. Einstad, I. Saltvedt, S. Lydersen, M.H. Ursin, R. Munthe-Kaas, H. Ihle-Hansen, A.-B. Knapskog, T. Askim, M.K. Beyer, H. Næss, Y.M. Seljeseth, H. Ellekjær, P. Thingstad, Associations between post-stroke motor and cognitive function: a cross-sectional study, BMC Geriatr. 21 (2021) 103. https://doi.org/10.1186/s12877-021-02055-7.

[41] V.M. Vargas, A.M. Durán-Rosal, D. Guijo-Rubio, P.A. Gutiérrez, C. Hervás-Martínez, Generalised triangular distributions for ordinal deep learning: Novel proposal and optimisation, Inf. Sci. 648 (2023) 119606. https://doi.org/10.1016/j.ins.2023.119606.

[42] Y. Gorishniy, I. Rubachev, V. Khrulkov, A. Babenko, Revisiting deep learning models for tabular data, Advances in Neural Information Processing Systems 34 (2021) 18932–18943.

[43] C.G. Maté, The iBoxplot: A visual tool based on lows and highs to support decision-making and acquire knowledge, Knowl.-Based Syst. 328 (2025) 114000. https://doi.org/10.1016/j.knosys.2025.114000.

[44] T.F. Tan, K. Elangovan, J. Ong, N. Shah, J. Sung, T.Y. Wong, L. Xue, N. Liu, H. Wang, C.F. Kuo, S. Chesterman, Z.K. Yeong, D.S.W. Ting, A proposed S.C.O.R.E. evaluation framework for large language models: Safety, consensus, objectivity, reproducibility and explainability, arXiv (2024). https://arxiv.org/abs/2407.07666.

[45] Y.H. Ke, L. Jin, K. Elangovan, J. Ong, T.F. Tan, P. Rajpurkar, N. Liu, D.S.W. Ting, Retrieval augmented generation for 10 large language models and its generalizability in assessing medical fitness, Npj Digit. Med. 8 (2025) 187. https://doi.org/10.1038/s41746-025-01519-z.

[46] Y. Tian, J. Li, N. Chen, G. Li, Z. Xu, H. Zhu, Y. Li, Z. Zhu, Heterogeneity-aware multi-modal physiological signal fusion strategy based on combined contrastive learning for emotion recognition, Neural Netw. 200 (2026) 108818. https://doi.org/10.1016/j.neunet.2026.108818.

[47] J. Lee, J. Jeon, K. Sohn, B. Ham, Mixture of zero-cost proxies for training-free network architecture search, Pattern Recognit. 180 (2026) 114211. https://doi.org/10.1016/j.patcog.2026.114211.

[48] Z. Zhu, F. Zhou, S. Huang, T. Li, EKCA-Cap: Improving dense captioning via external knowledge and context awareness, Neurocomputing 635 (2025) 129867. https://doi.org/10.1016/j.neucom.2025.129867.

[49] Z. Wang, J. Wang, J. Guo, A. Dove, K. Arfanakis, X. Qi, D.A. Bennett, W. Xu, Association of Motor Function With Cognitive Trajectories and Structural Brain Differences, Neurology 101 (2023) e1718–e1728. https://doi.org/10.1212/WNL.0000000000207745.

[50] H. Wang, J. Guo, Y. Zhang, Z. Fu, Y. Yao, Closed-loop rehabilitation of upper-limb dyskinesia after stroke: from natural motion to neuronal microfluidics, J. NeuroEngineering Rehabil. 22 (2025) 87. https://doi.org/10.1186/s12984-025-01617-9.

[51] S. Das, Y. Ge, Y. Guo, S. Rajwal, J. Hairston, J. Powell, D. Walker, S. Peddireddy, S. Lakamana, S. Bozkurt, M. Reyna, R. Sameni, Y. Xiao, S. Kim, R. Chandler, N. Hernandez, D. Mowery, R. Wightman, J. Love, A. Spadaro, J. Perrone, A. Sarker, Two-Layer Retrieval-Augmented Generation Framework for Low-Resource Medical Question Answering Using Reddit Data: Proof-of-Concept Study, J. Med. Internet Res. 27 (2025) e66220. https://doi.org/10.2196/66220.

[52] X. Li, F. Chen, W. Wu, L. Zeng, Z. Gu, Y. Zhang, Pyramid graph neural network knowledge distillation with pre-trained language model for medical question answering, Eng. Appl. Artif. Intell. 176 (2026) 114841. https://doi.org/10.1016/j.engappai.2026.114841.